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Kishimoto et al.

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(54) SHEET POST-PROCESSING DEVICE

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B65H 31/36 (2006.01)
B65H 29/52 (2006.01)
B65H 31/26 (2006.01)

(52) **U.S. Cl.**
CPC ***B65H 31/36*** (2013.01); ***B65H 29/52*** (2013.01); ***B65H 31/26*** (2013.01); *B65H 2801/27* (2013.01)

(58) **Field of Classification Search**
CPC B65H 29/52; B65H 31/36; B65H 31/26
See application file for complete search history.

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(74) *Attorney, Agent, or Firm* — Stein IP LLC

(57) **ABSTRACT**

A sheet post-processing device includes a discharge port, a discharge tray, a first sheet pressing member, a second sheet pressing member, and a holding member. The holding member has a first holding portion that holds the first sheet pressing member, and a second holding portion that holds the second sheet pressing member. The first holding portion selectively holds the first sheet pressing member at a first pressing position for pressing a sheet and a first retreat position retreated to above a discharge route. The second holding portion selectively holds the second sheet pressing member at a second pressing position for pressing the sheet and a second retreat position retreated to above the discharge route.

7 Claims, 11 Drawing Sheets

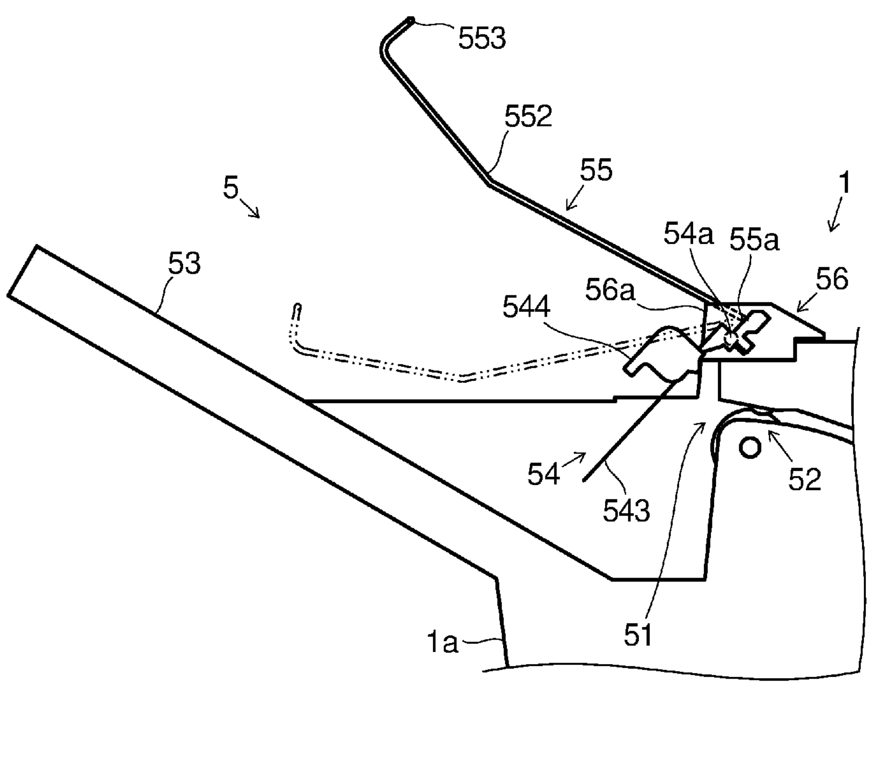
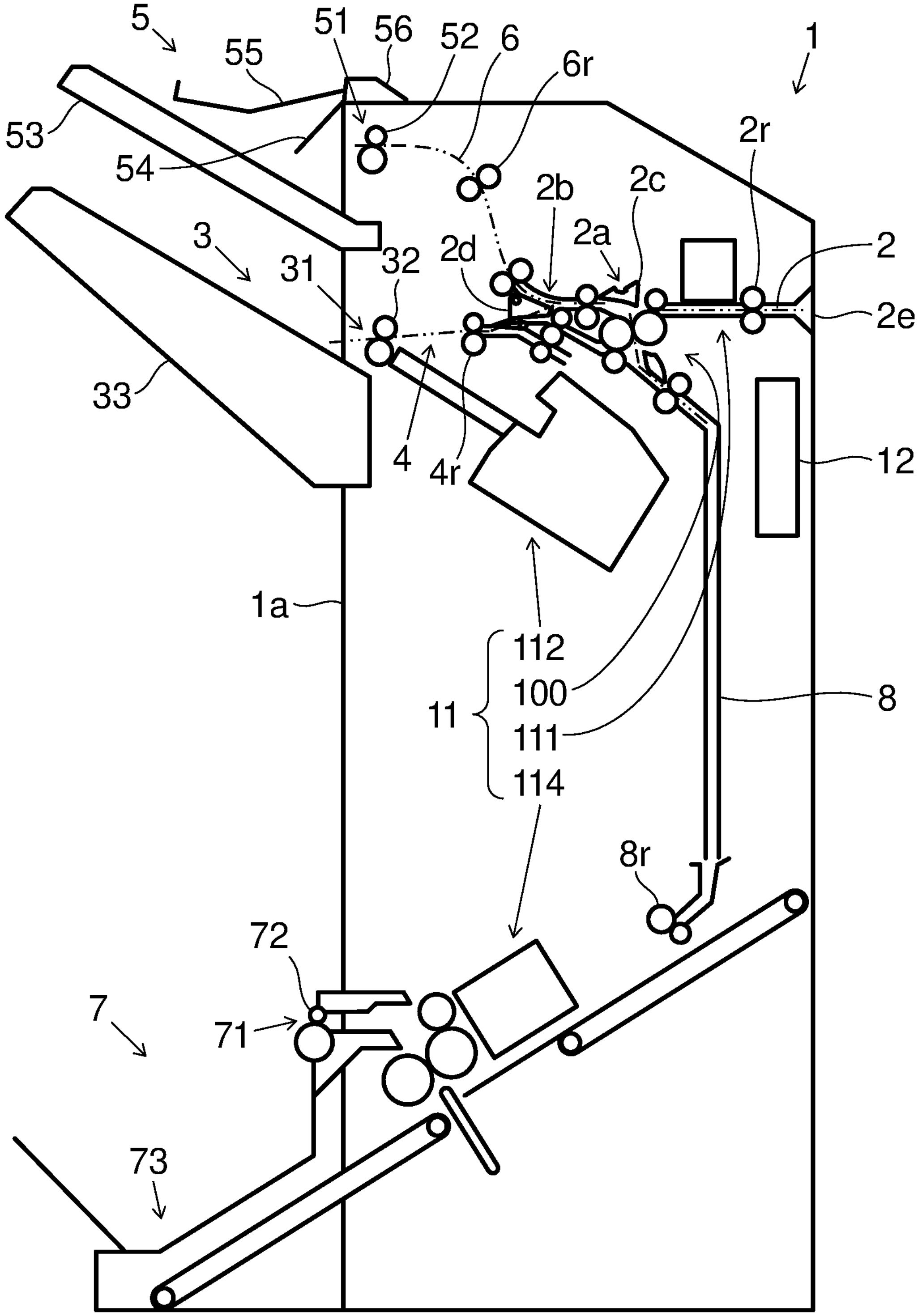


FIG.1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51
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54
2r
2b
2c
2a
2
2d
3
31
32
2e
33
4
4r
12
1a
112
100
11
111
114
8
8r
72
7
71
73

FIG.2A

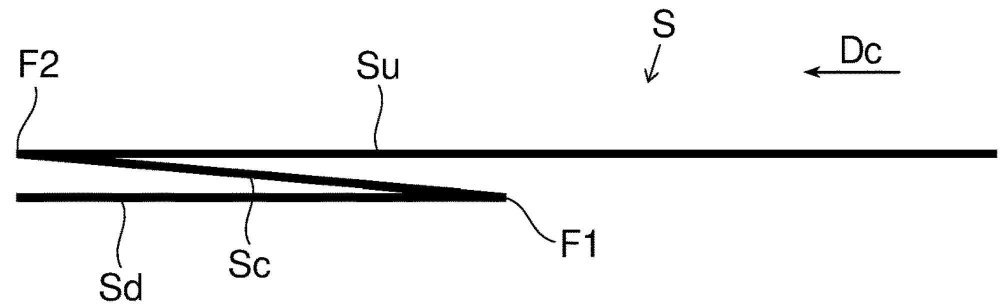


FIG.2B

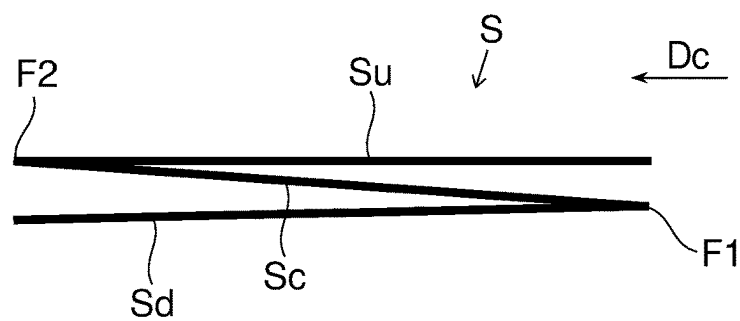


FIG.2C

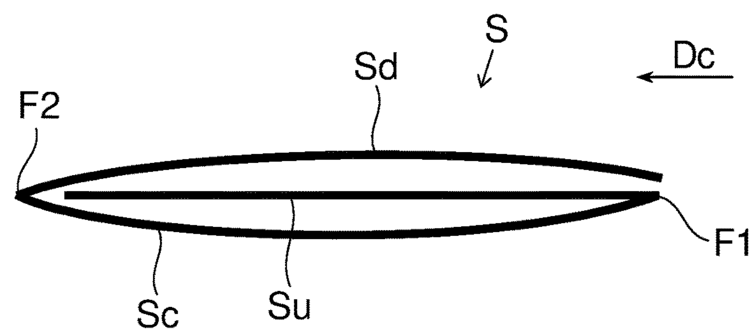
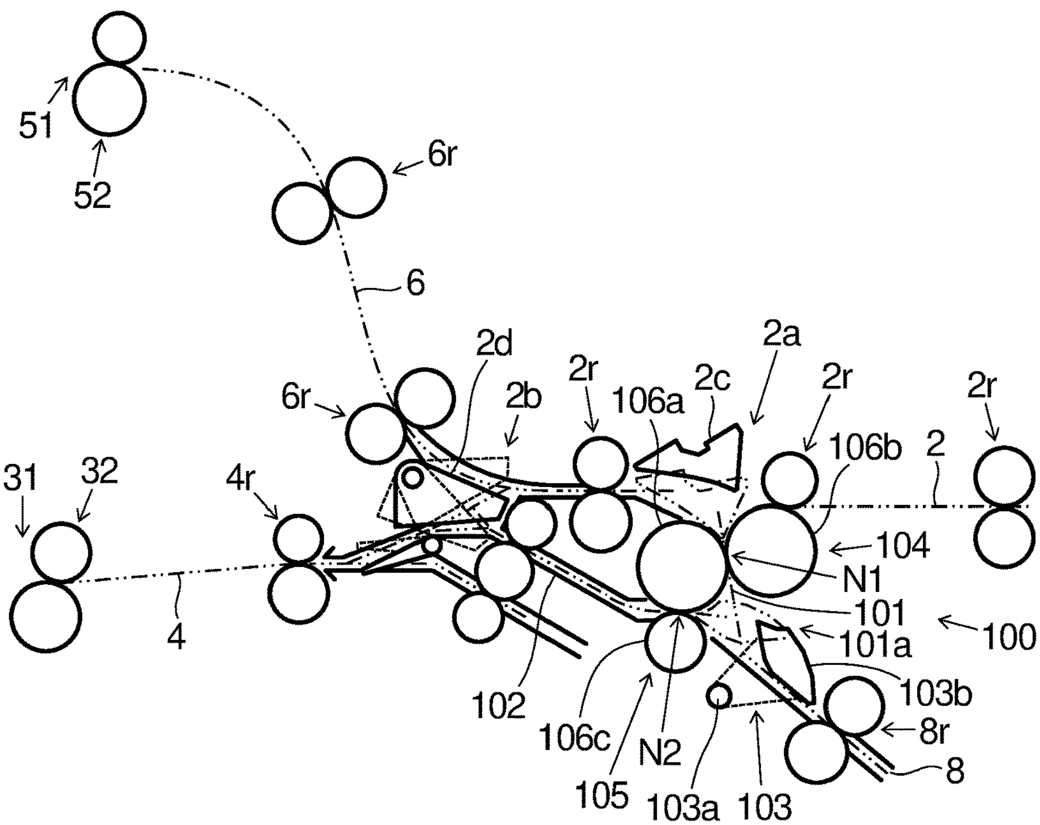
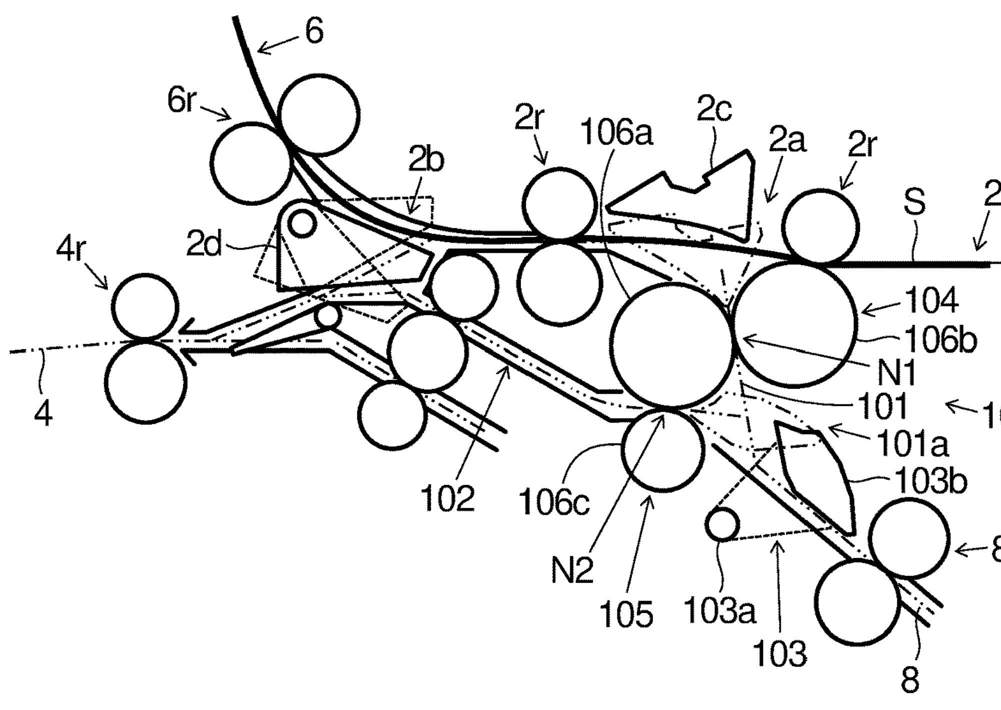


FIG.3



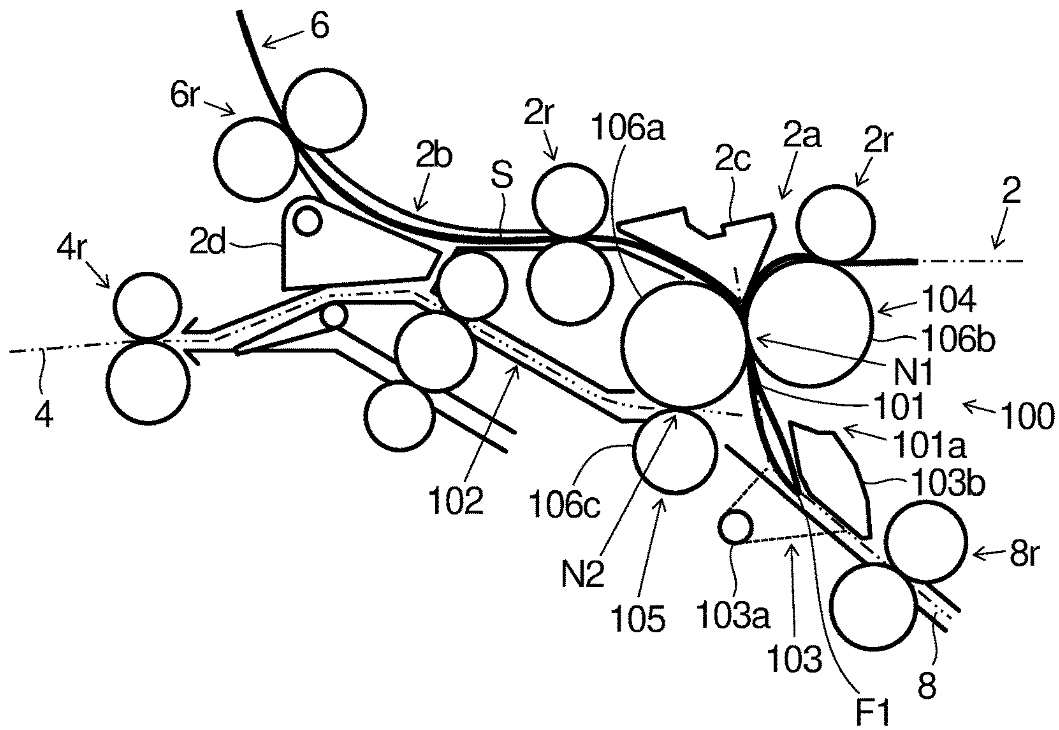
51
52
6r
6
2d
2b
2r
106a
2c
2a
106b
2
31
32
4r
104
N1
101
101a
100
4
102
103b
8r
8
106c
N2
105
103a
103

FIG.4



6
6r
2b
2r
106a
2c
2a
S
2
2d
4r
104
106b
N1
101
101a
100
4
102
106c
103b
8r
N2
105
103a
103
8

FIG.5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2r
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4r
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106b
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101a
103b
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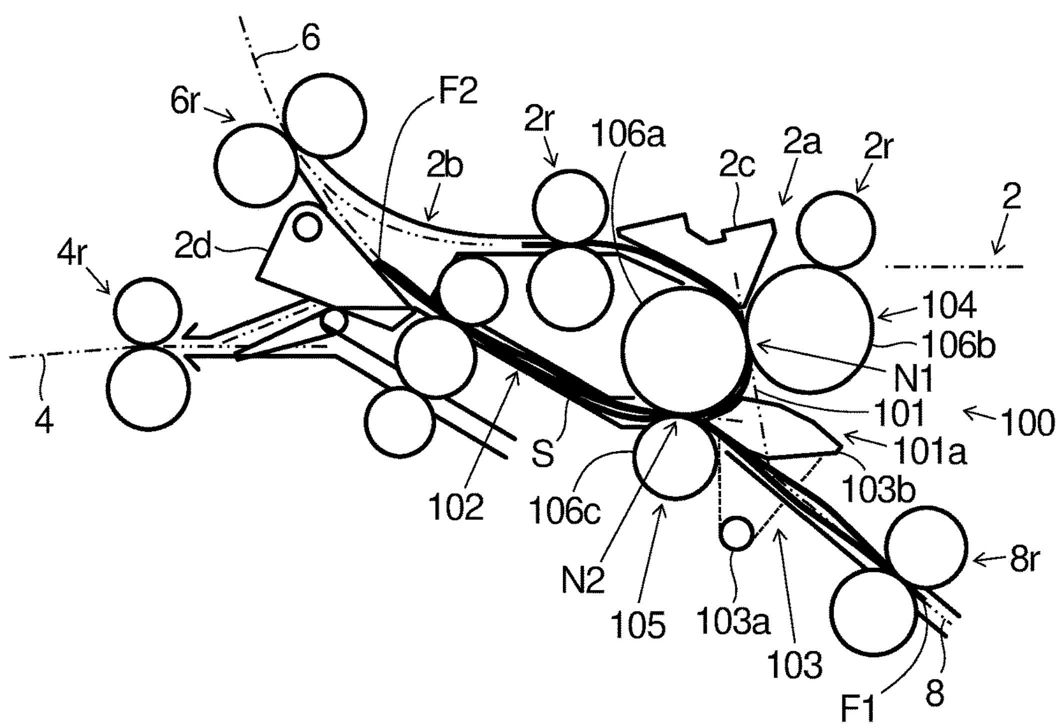
106c
8r
N2
105
103a
103
8
F1

FIG.6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2r
2b
2c
2
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2d
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106b
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101a
S
103b
102
106c
8r
N2
105
103a
103
F1
8

FIG.7

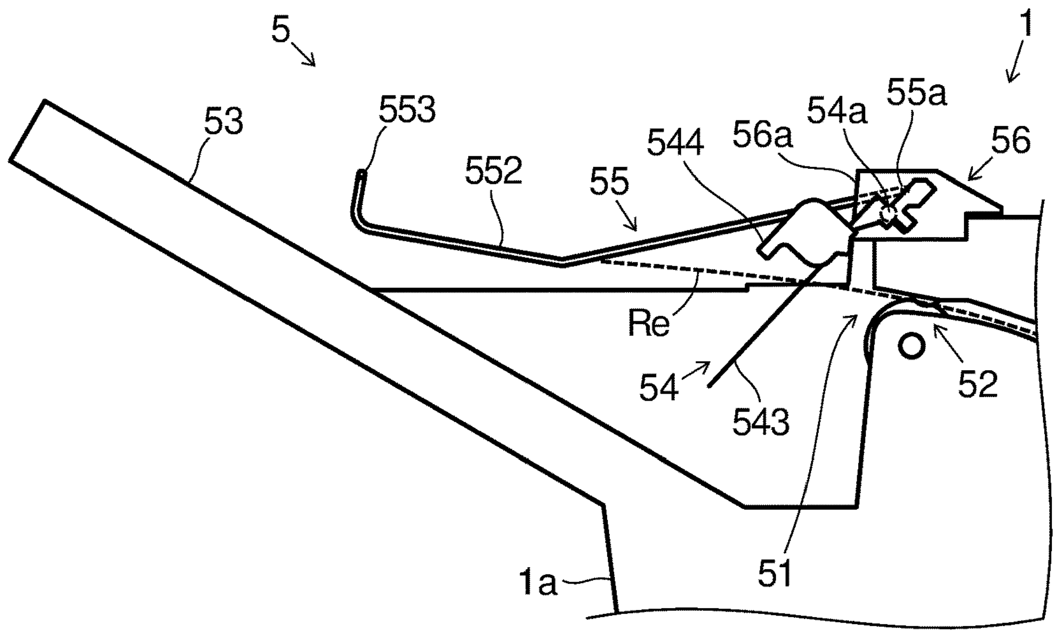
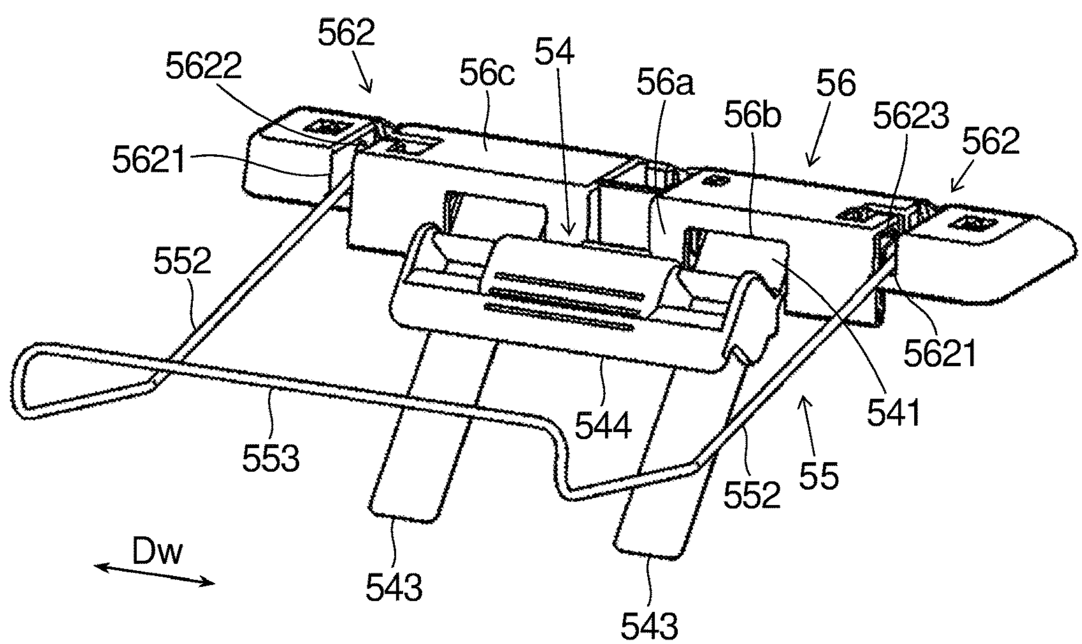

5
1
553
53
55a
54a
544
56a
56
552
55
Re
54
543
52
51
1a

FIG.8


562
5622
56c
54
56a
56
56b
5623
562
5621
552
5621
541
544
553
55
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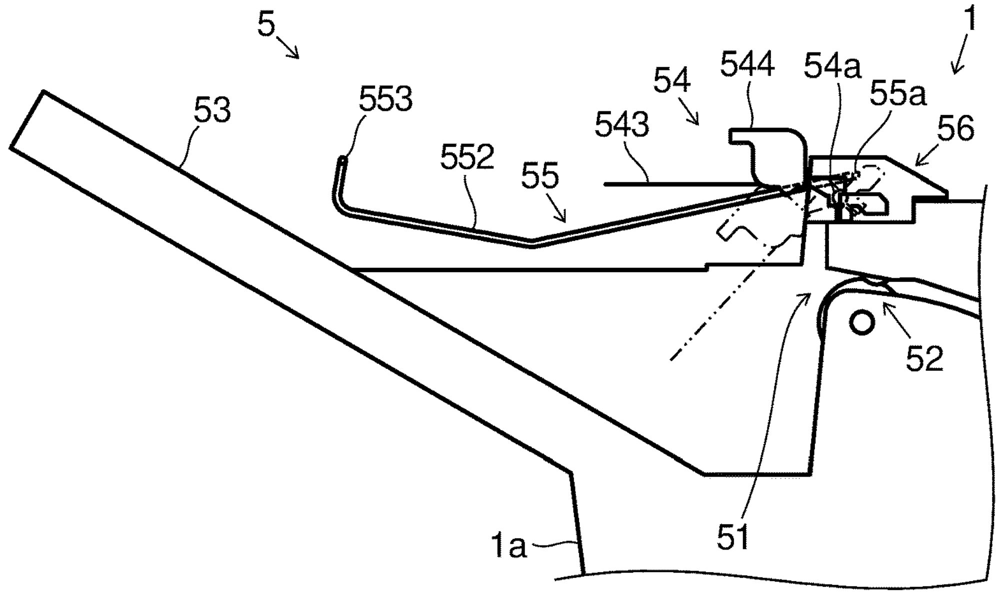
Dw
543
543

FIG.9



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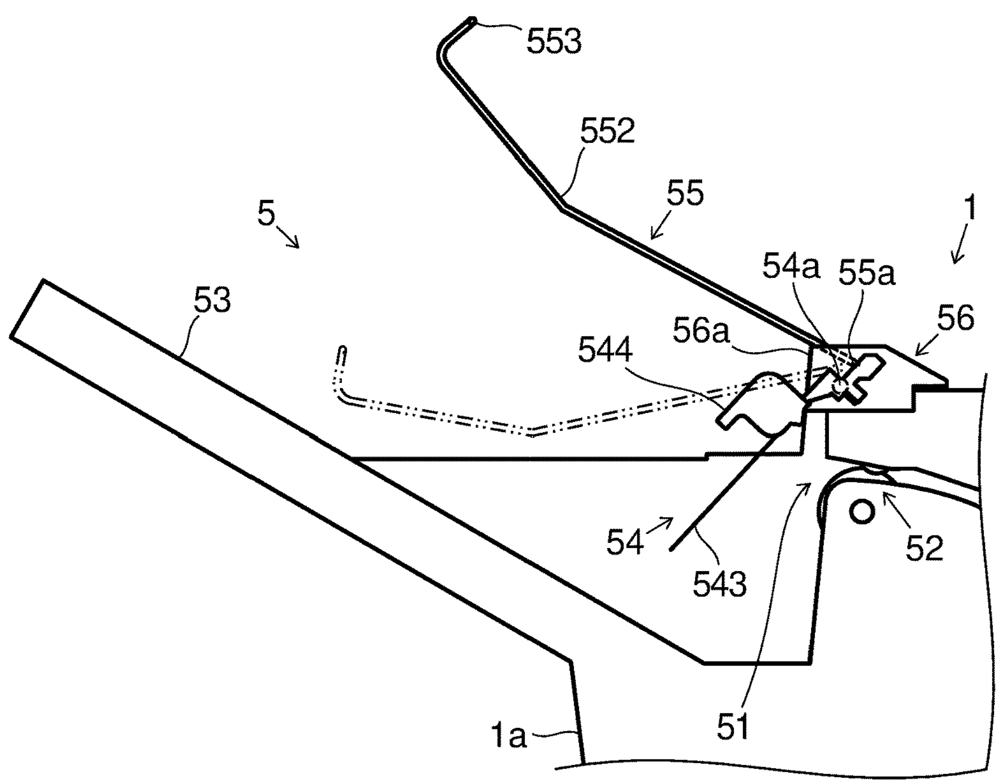
1
53
553
552
55
54
544
54a
55a
543
56
52
51
1a

FIG.10



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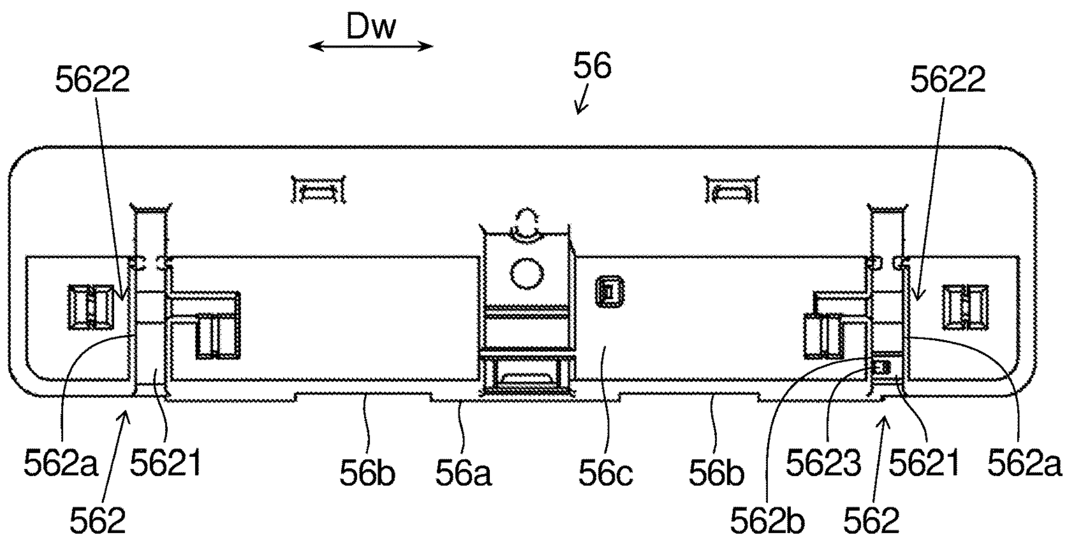
5
1
53
54a
55a
56
56a
544
54
543
52
51
1a

FIG.11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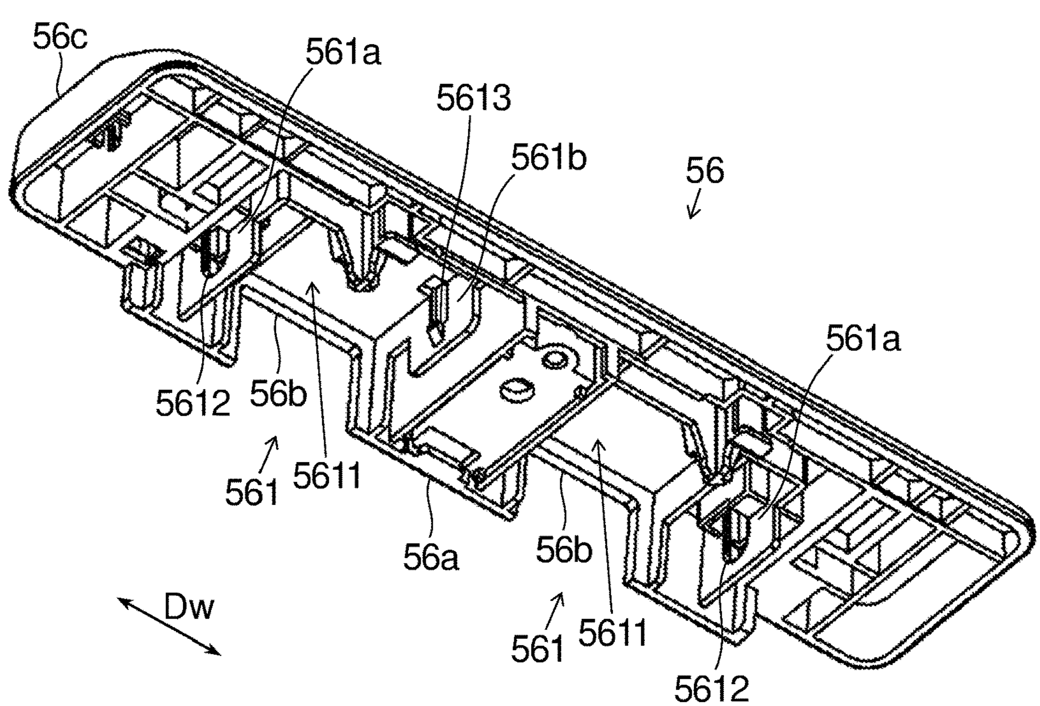
5621
562a

FIG.12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561a
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56b
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5611
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56b
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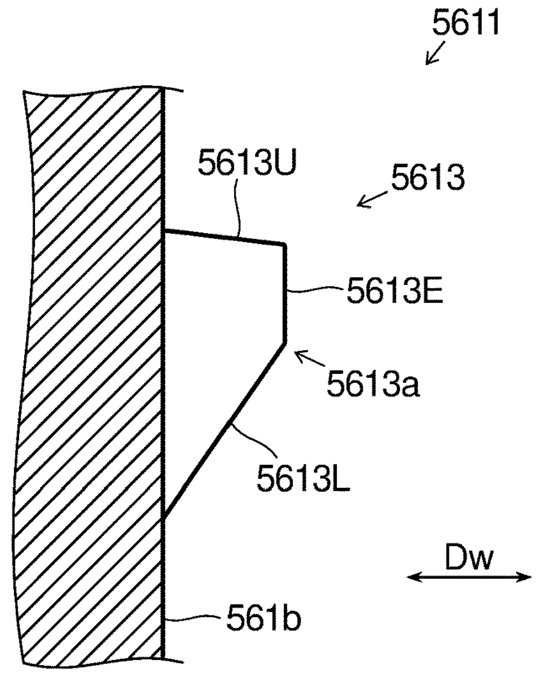
5612
Dw

FIG.13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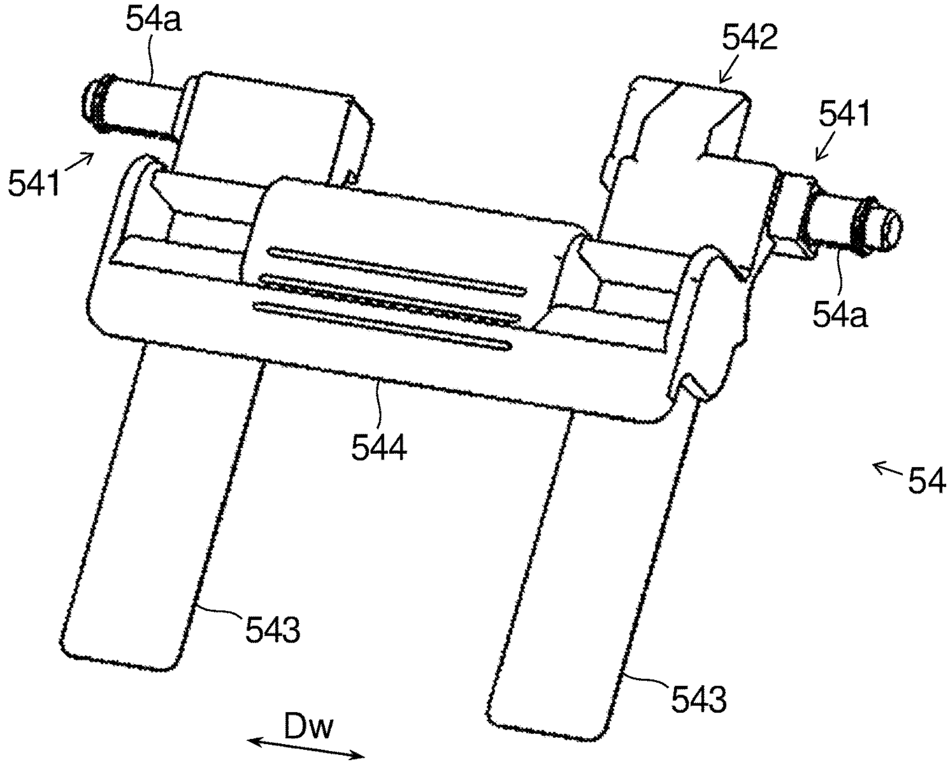
Dw
561b

FIG.14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542
541
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54a
544
54
543
543
Dw

FIG.15

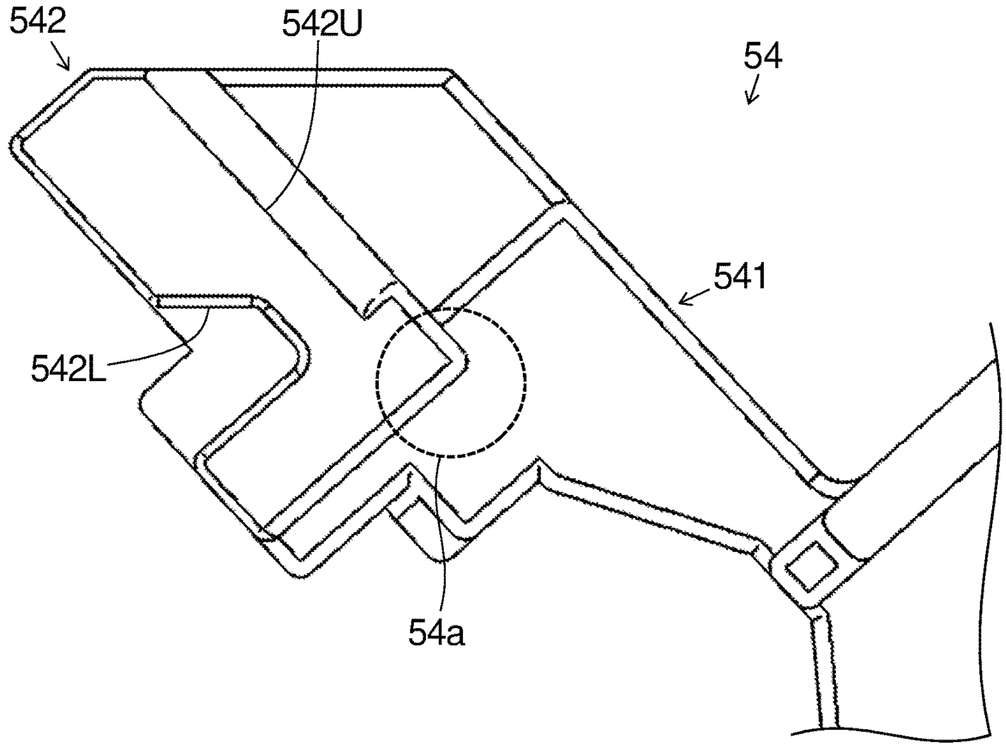
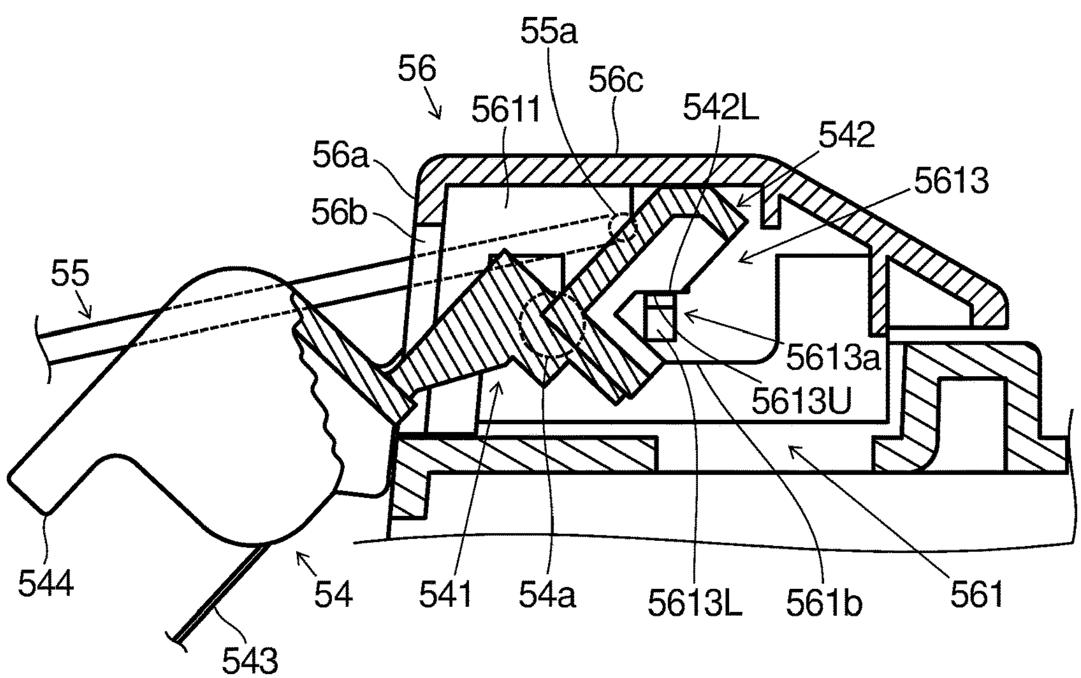

542
542U
54
541
542L
54a

FIG.16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561b
561
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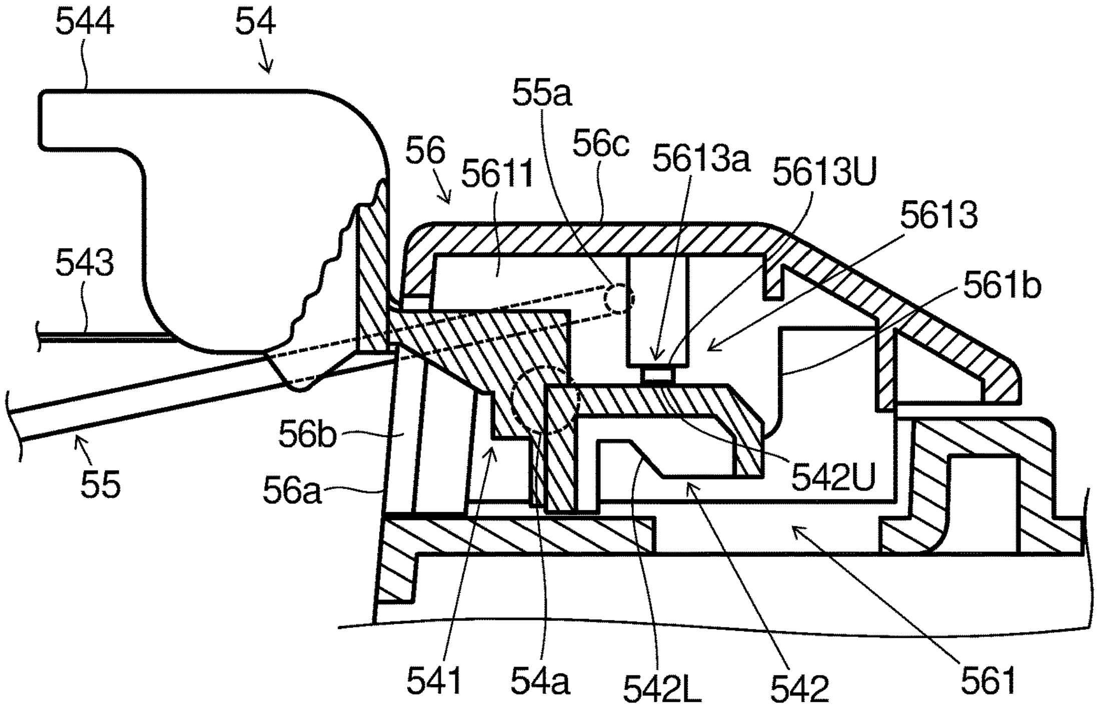
FIG.17



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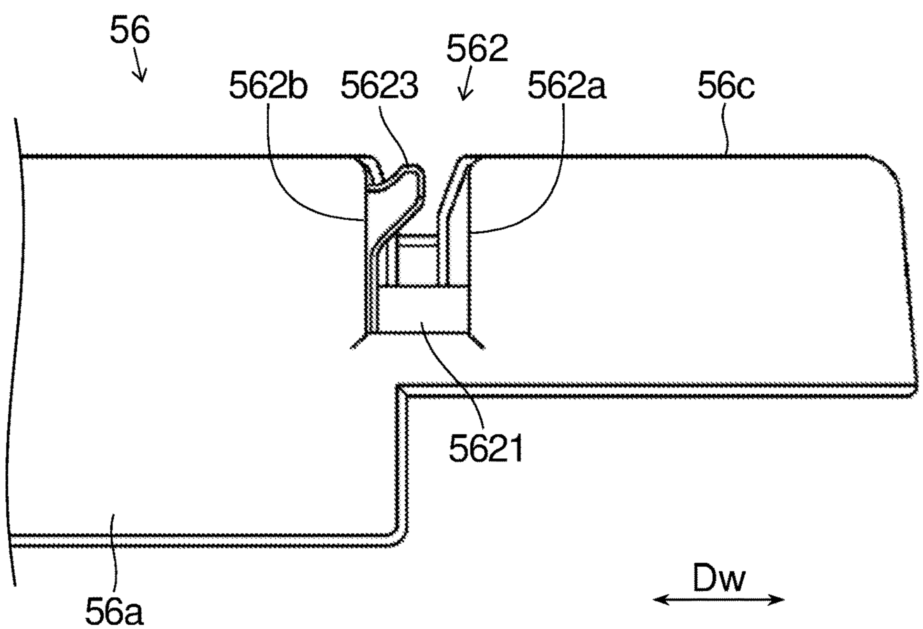
542L
542
561

FIG.18



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562b
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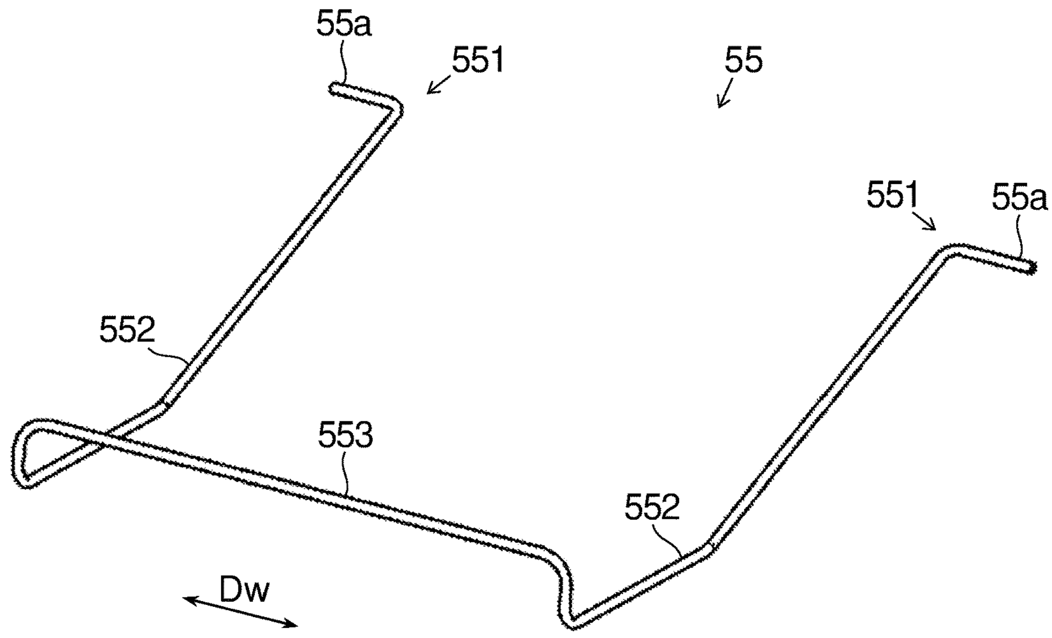
562a
56c
5621
56a
Dw

FIG.19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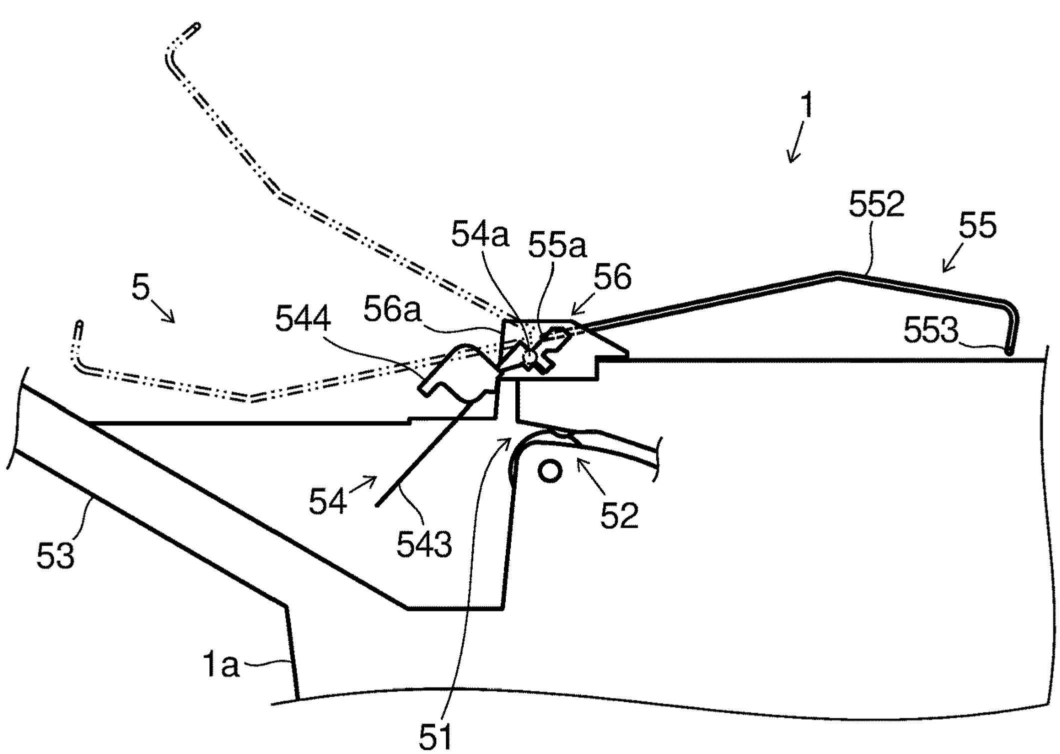
55a
552
553
552
Dw

FIG.20


1
552
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54a
55a
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5
544
56a
553
54
52
543
53
1a
51

SHEET POST-PROCESSING DEVICE

INCORPORATION BY REFERENCE

This application is based on and claims the benefit of priority from Japanese Patent Application No. 2023-028738 filed on Feb. 27, 2023, the contents of which are hereby incorporated by reference.

BACKGROUND

The present disclosure relates to a sheet post-processing device that performs post-processing with respect to a sheet having had an image formed thereon by an image forming apparatus.

A known sheet post-processing device performs folding processing of forming a fold in a sheet having had an image formed thereon by an image forming apparatus such as a copier, a printer, or the like. In a case where sheets subjected to such folding processing are discharged to be stacked on a discharge tray, the folded parts of the sheets make the stack of the sheets partly thick, making it difficult to stack a large number of such sheets in an orderly manner. Measures have conventionally been taken to deal with this inconvenience.

SUMMARY

According to one aspect of the present disclosure, a sheet post-processing device includes a discharge port, a discharge tray, a first sheet pressing member, a second sheet pressing member, and a holding member. The discharge port has a pair of discharge rollers that discharge a sheet. On the discharge tray, the sheet discharged through the discharge port is stacked. The first sheet pressing member is attached above the pair of discharge rollers, and presses an upstream part, in a sheet discharge direction, of the sheet stacked on the discharge tray. The second sheet pressing member is attached above the pair of discharge rollers, and presses a downstream part, in the sheet discharge direction, of the sheet stacked on the tray. The holding member holds the first sheet pressing member and the second sheet pressing member. The holding member has a first holding portion and a second holding portion. The first holding portion holds the first sheet pressing member such that the first sheet pressing member is swingable to a first pressing position, at which the first sheet pressing member extends, intersecting a discharge route of the sheet, downward from above the pair of discharge rollers toward the discharge tray to press the sheet, and to a first retreat position retreated to above the discharge route of the sheet. The second holding portion holds the second sheet pressing member such that the second sheet pressing member is swingable to a second pressing position, at which the second sheet pressing member extends, intersecting the discharge route of the sheet, downward from above the pair of discharge rollers toward the discharge tray to press the sheet, and to a second retreat position retreated to above the discharge route of the sheet. The first holding portion selectively holds the first sheet pressing member at the first pressing position and the first retreat position. The second holding portion selectively holds the second sheet pressing member at the second pressing position and the second retreat position.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic sectional front view of a sheet post-processing device according to an embodiment of the present disclosure.

FIG. **2**A is a schematic front view of a Z-folded sheet.

FIG. **2**B is a schematic front view of an outward triple-folded sheet.

FIG. **2**C is a schematic front view of an inward triple-folded sheet.

FIG. **3** is a partial sectional front view showing a sheet folding portion of the sheet post-processing device and its vicinity shown in FIG. **1**.

FIG. **4** is a sectional front view showing the sheet folding portion and its vicinity shown in FIG. **3**, illustrating a first stage in the course of inward triple-folding processing for a sheet.

FIG. **5** is a sectional front view showing the sheet folding portion and its vicinity shown in FIG. **3**, illustrating a second stage in the course of the inward triple-folding processing for a sheet.

FIG. **6** is a sectional front view showing the sheet folding portion and its vicinity shown in FIG. **3**, illustrating a third stage in the course of the inward triple-folding processing for a sheet.

FIG. **7** is a partial sectional front view showing a second sheet discharge portion of the sheet post-processing device and its vicinity shown in FIG. **1**.

FIG. **8** is a perspective view showing a first sheet pressing member, a second sheet pressing member, and a holding member of the second sheet discharge portion shown in FIG. **7**.

FIG. **9** is a partial sectional front view showing the second sheet discharge portion and its vicinity shown in FIG. **7**, with the first sheet pressing member positioned at a retreat position.

FIG. **10** is a partial sectional front view showing the second sheet discharge portion and its vicinity shown in FIG. **7**, with the second sheet pressing member positioned at a retreat position.

FIG. **11** is a top view of the holding member of the second sheet discharge portion shown in FIG. **7**.

FIG. **12** is a perspective view, as seen from below, of the holding member of the second sheet discharge portion shown in FIG. **7**.

FIG. **13** is a side view of a first protruding portion of the holding member shown in FIG. **12**, as seen from a sheet discharge direction.

FIG. **14** is a perspective view of the first sheet pressing member of the second sheet discharge portion shown in FIG. **7**.

FIG. **15** is a partial rear view of the first sheet pressing member shown in FIG. **14**.

FIG. **16** is a partial sectional front view of the first sheet pressing member (a first pressing position) and the holding member of the second sheet discharge portion shown in FIG. **7**.

FIG. **17** is a partial sectional front view of the first sheet pressing member (a first retreat position) and the holding member of the second sheet discharge portion shown in FIG. **7**.

FIG. **18** is a partial side view showing a second holding portion of the holding member of the second sheet discharge portion and its vicinity shown in FIG. **7**.

FIG. **19** is a perspective view of the second sheet pressing member of the second sheet discharge portion shown in FIG. **7**.

FIG. **20** is a partial sectional front view showing the second discharge portion and its vicinity shown in FIG. **7**, with the second sheet pressing member positioned at a third retreat position.

DETAILED DESCRIPTION

An embodiment of the present disclosure will be described below with reference to the accompanying drawings. It should be understood, however, that the present disclosure is not limited to what is specifically described below.

FIG. **1** is a schematic sectional front view of a sheet post-processing device **1** according to an embodiment. The sheet post-processing device **1** is coupled to a side surface of an image forming apparatus (unillustrated), for example, so as to be attachable and detachable. The sheet post-processing device **1** performs post-processing with respect to a sheet having had an image formed (printed) thereon by the image forming apparatus.

As shown in FIG. **1**, the sheet post-processing device **1** includes a sheet introduction path **2**, a first sheet discharge portion **3**, a first sheet discharge path **4**, a second sheet discharge portion **5**, a second sheet discharge path **6**, a third sheet discharge portion **7**, a third sheet discharge path **8**, a post-processing portion **11**, and a post-processing control portion **12**.

The sheet introduction path **2** has a sheet introduction port **2***e*. The sheet introduction port **2***e* is provided, and opened, in a side surface of the sheet post-processing device **1**, the side surface facing the image forming apparatus (unillustrated). A sheet conveyed from the image forming apparatus toward the sheet post-processing device **1** passes through the sheet introduction port **2***e* and the sheet introduction path **2** to be introduced into the sheet post-processing device **1**.

The sheet introduction path **2** substantially horizontally extends from the sheet introduction port **2***e* to a sheet folding portion **100**, which will be described later, in a direction (a leftward direction in FIG. **1**) away from the image forming apparatus. In this description, a direction from the sheet introduction port **2***e* toward an inside of the sheet post-processing device **1** is referred to as a sheet conveyance direction along the sheet introduction path **2**. The sheet introduction port **2***e* is located at an upstream end of the sheet introduction path **2** in the sheet conveyance direction. In the sheet introduction path **2**, pairs of feed rollers **2***r* are disposed.

The pairs of feed rollers **2***r* convey, on the sheet introduction path **2**, a sheet introduced through the sheet introduction port **2***e* into the sheet post-processing device **1**, toward a downstream side in the sheet conveyance direction.

The sheet introduction path **2** has a first branching portion **2***a* and a second branching portion **2***b*. The first branching portion **2***a* is located upstream of the second branching portion **2***b* with respect to the sheet conveyance direction along the sheet introduction path **2**. The second branching portion **2***b* is located at a downstream end part of the sheet introduction path **2** in the sheet conveyance direction.

At the first branching portion **2***a*, a first switching guide **2***c* is disposed. The first switching guide **2***c* switches a conveyance direction of a sheet conveyed on the sheet introduction path **2** from a side of the sheet introduction port **2***e* toward the downstream side in the sheet conveyance direction between a direction leading to the first sheet discharge portion **3** and the second sheet discharge portion **5** and a direction leading to the third sheet discharge portion **7**.

At the second branching portion **2***b*, a second switching guide **2***d* is disposed. The second switching guide **2***d* switches the conveyance direction of the sheet conveyed on the sheet introduction path **2** from the side of the sheet introduction port **2***e* toward the downstream side in the sheet conveyance direction between a direction leading to the first sheet discharge portion **3** and a direction leading to the second sheet discharge portion **5**.

The first sheet discharge portion **3** is provided on a side surface **1***a* of the sheet post-processing device **1**, the side surface **1***a* being located opposite the side surface of the sheet post-processing device **1** that faces the image forming apparatus. The first sheet discharge portion **3** has a first discharge port **31**, a pair of first discharge rollers **32**, and a first discharge tray **33**.

The first discharge port **31** is located at a downstream end of the first sheet discharge path **4** in the sheet conveyance direction. The pair of first discharge rollers **32** are disposed at the first discharge port **31**. The first discharge tray **33** is located downstream of the first discharge port **31** in the sheet conveyance direction. A sheet having been conveyed along the first sheet discharge path **4** to reach the first discharge port **31** is discharged, by the pair of first discharge rollers **32**, through the first discharge port **31** onto the first discharge tray **33**. The first discharge tray **33** is one of final discharge destinations of sheets subjected to the post-processing performed by the sheet post-processing device **1**.

The first sheet discharge path **4** is continuous with the downstream end of the sheet introduction path **2** in the sheet conveyance direction, and substantially horizontally extends to the first sheet discharge portion **3** in the direction (the leftward direction in FIG. **1**) away from the image forming apparatus. In this description, a direction (the leftward direction in FIG. **1**) from the inside of the sheet post-processing device **1** toward the first discharge portion **3** is referred to as a sheet conveyance direction along the first sheet discharge path **4**. In the first sheet discharge path **4**, pairs of feed rollers **4***r* are disposed. The pairs of feed rollers **4***r* convey a sheet having passed through the sheet introduction path **2** to reach the second branching portion **2** toward a downstream side on the first sheet discharge path **4** in the sheet conveyance direction, that is, toward the first sheet discharge portion **3**.

The second sheet discharge portion **5** is provided on the side surface **1***a* of the sheet post-processing device **1** located opposite the side surface thereof facing the image forming apparatus, above the first sheet discharge portion **3**. The second sheet discharge portion **5** has a second discharge port (a discharge port) **51**, a pair of second discharge rollers (a pair of discharge rollers) **52**, and a second discharge tray (a discharge tray) **53**.

The second discharge port **51** is located at a downstream end of the second sheet discharge path **6** in the sheet conveyance direction. The pair of second discharge rollers **52** are disposed at the second discharge port **51**. The second discharge tray **53** is located downstream of the second discharge port **51** in the sheet conveyance direction. A sheet having been conveyed through the second sheet discharge path **6** to reach the second discharge port **51** is discharged, by the pair of second discharge rollers **52**, through the second discharge port **51** onto the second discharge tray **53**. The second discharge tray **53** is one of the final discharge destinations of sheets subjected to the post-processing performed by the sheet post-processing device **1**.

The second sheet discharge path **6** branches off from the second branching portion **2** located at a downstream part of the sheet introduction path **2** in the sheet conveyance direction, and the second sheet discharge path **6** extends not only laterally in the direction (the leftward direction in FIG. **1**) away from the image forming apparatus but also in an upward direction to the second sheet discharge portion **5**. In this description, a direction (an upper-leftward direction in

FIG. **1**) from the second branching portion **2***b* toward the second sheet discharge portion **5** is referred to as a sheet conveyance direction along the second sheet discharge path **6**. In the second sheet discharge path **6**, pairs of feed rollers **6***r* are disposed. The pairs of feed rollers **6***r* convey a sheet having passed through the sheet introduction path **2** to reach the second branching portion **2** toward a downstream side on the second sheet discharge path **6** in the sheet conveyance direction, that is, toward the second sheet discharge portion **5**.

The third sheet discharge portion **7** is provided on the side surface **1***a* of the sheet post-processing device **1** located opposite the side surface thereof facing the image forming apparatus, below the first sheet discharge portion **3**. In other words, the third sheet discharge portion **7** is disposed near a bottom part of the sheet post-processing device **1**. The third sheet discharge portion **7** has a third discharge port **71**, a pair of third discharge rollers **72**, and a third discharge tray **73**.

The third discharge port **71** is located at a downstream end of the third sheet discharge path **8** in the sheet conveyance direction. The pair of third discharge rollers **72** are disposed at the third discharge port **71**. The third discharge tray **73** is located downstream of the third discharge port **71** in the sheet conveyance direction. A sheet having been conveyed through the third sheet discharge path **8** to reach the third discharge port **71** is discharged, by the pair of third discharge rollers **72**, through the third discharge port **71** onto the third discharge tray **73**. The third discharge tray **73** is one of the final discharge destinations of sheets subjected to the post-processing performed by the sheet post-processing device **1**.

The third sheet discharge path **8** branches off from the first branching portion **2***a* located at the downstream part of the sheet introduction path **2** in the sheet conveyance direction, and extends in a downward direction to the third sheet discharge portion **7**. In this description, a direction from the first branching portion **2***a* toward the third sheet discharge portion **7** is referred to as a sheet conveyance direction along the third sheet discharge path **8**. In the third sheet discharge path **8**, pairs of feed rollers **8***r* are disposed. The pairs of feed rollers **8***r* convey a sheet having passed through the sheet introduction path **2** to reach the first branching portion **1** toward a downstream side on the third sheet discharge path **8** in the sheet conveyance direction, that is, toward the third sheet discharge portion **7**.

The post-processing portion **11** performs predetermined post-processing with respect to a sheet having been subjected to image formation performed by the image forming apparatus and introduced into the sheet post-processing device **1**. The post-processing portion **11** includes a punching portion **111**, a stapling portion **112**, a sheet folding portion **100**, and a book-binding portion **114**.

The punching portion **111** is disposed in the sheet introduction path **2**, closely downstream of the sheet introduction port **2***e*. The punching portion **111** is capable of performing punching processing with respect to a sheet conveyed on the sheet introduction path **2**, thereby to form a punch hole in the sheet.

The stapling portion **112** is disposed below the first sheet discharge path **4**, in the vicinity of the first sheet discharge portion **3**. The stapling portion **112** is capable of performing stapling processing (binding processing) with respect to a bundle of sheets formed by stacking together a plurality of sheets, thereby to bind the bundle of sheets.

The sheet folding portion **100** is disposed, with respect to the sheet conveyance direction along the sheet introduction path **2**, downstream of the punching portion **111**, upstream of the stapling portion **112**. The sheet folding portion **100** is capable of performing folding processing with respect to a single sheet, thereby to make a fold in the single sheet.

The sheet folding portion **100** is capable of performing, with respect to a single sheet, processing such as double-folding, Z-folding, outward triple-folding, inward triple-folding, quadruple-folding, etc. FIGS. **2**A, **2**B, and **2**C are schematic front views of a Z-folded sheet S, an outward triple-folded sheet S, and an inward triple-folded sheet S, respectively.

Z-folding is a manner of folding in which, as shown in FIG. **2**A, for example, a downstream part of a sheet S with respect to the sheet conveyance direction Dc along the sheet introduction path **2** is formed into a Z-shape as seen from a sheet width direction (the depth direction of the plane of FIG. **2**A) orthogonal to the sheet conveyance direction. In Z-folding, a downstream part Sd of the sheet S downstream of a first fold F**1** in the sheet conveyance direction Dc and an upstream part Su of the sheet S upstream of a second fold F**2** face each other in an up-down direction across a middle part Sc of the sheet S between the two folds. In the sheet conveyance direction Dc, the downstream part Sd and the middle part Sc of the sheet S are approximately equal in length, but are shorter than the upstream part Su in length.

Outward triple-folding is a manner of folding in which, as shown in FIG. **3**, for example, an entire sheet S is formed into a Z-shape as seen from the sheet width direction (the depth direction of the plane of FIG. **2**B). In outward triple folding, a downstream part Sd of the sheet S downstream of a first fold F**1** in the sheet conveyance direction Dc and an upstream part Su of the sheet S upstream of a second fold F**2** face each other in the up-down direction across a middle part Sc of the sheet S between the two folds. In the sheet conveyance direction Dc, the downstream part Sd, the middle part Sc, and the upstream part Su of the sheet S are approximately equal in length.

In inward triple-folding, as shown in FIG. **2**C, for example, an upstream part Su of a sheet S upstream of a first fold F**1** in the sheet conveyance direction Dc and a downstream part Sd of the sheet S downstream of a second fold F**2** face each other in the up-down direction and make surface contact with each other at one side of (in FIG. **2**C, above) the plane of a middle part Sc of the sheet S between the two folds.

In this description, a sheet having been subjected to the folding processing may be referred to as a "folded sheet."

The book-binding portion **114** is disposed in a downstream part of the third sheet discharge path **8** in the sheet conveyance direction, in the vicinity of the third sheet discharge portion **7**. The book-binding portion **114** is capable of performing, with respect to a bundle of sheets formed by stacking a plurality of sheets, middle-folding processing and middle-binding processing in which the book-binding portion **114** folds and binds the bundle of sheets substantially at its middle, thereby to form a booklet.

The post-processing control portion **12** includes a CPU, an image processor, a storage, and other electronic circuits and electronic components (of which none is shown). The post-processing control portion **12** is communicably connected to a main control portion of the image forming apparatus (not shown). The post-processing control portion **12** receives instructions from the main control portion, and by means of the CPU and based on a control program and control data stored in the storage, controls operations of various components provided in the sheet post-processing device **1** so as to perform processing related to functions of the sheet conveyance device **1**. The sheet introduction path **2**, the first sheet discharge portion **3**, the first sheet discharge path **4**, the second sheet discharge portion **5**, the second sheet discharge path **6**, the third sheet discharge portion **7**, the third sheet discharge path **8**, and the post-processing portion **11** individually receive instructions from the post-processing control portion **12** to cooperate so as to perform post-processing on sheets. The functions of the post-processing control portion **12** may be assumed by the main control portion of the image forming apparatus.

Next, a configuration of the sheet folding portion **100** will be described with reference to FIG. **3**. FIG. **3** is a partial sectional front view showing the sheet folding portion **100** of the sheet post-processing device **1** and its vicinity shown in FIG. **1**. The sheet folding portion **100** includes a first sheet conveyance path **101**, a second sheet conveyance path **102**, and a folding blade **103**.

The first sheet conveyance path **101** is configured as an upstream part of the third sheet discharge path **8** in the sheet conveyance direction. That is, the first sheet conveyance path **101** branches off from the first branching portion **2***a* which is on the sheet introduction path **2**, and extends in a direction toward where the third sheet discharge portion **7** is located, in a downward direction in FIG. **3**. Along the first sheet conveyance path **101**, a sheet is conveyed.

The second sheet conveyance path **102** branches off from a folding branching portion **101***a* which is on the first sheet conveyance path **101**, and extends in a direction for intersecting the first sheet conveyance path **101**. The second sheet conveyance path **102** extends from the folding branching portion **101***a* toward a side-face-**1***a* side part of the sheet post-processing device **1** where the first sheet discharge path **4** is provided, in a leftward direction in FIG. **3**. Along the second sheet conveyance path **102**, a sheet is conveyed. A downstream end of the second sheet conveyance path **102** in the sheet conveyance direction merges with the second branching portion **2***b*.

The folding blade **103** is disposed in the vicinity of the folding branching portion **101***a* which is on the first sheet conveyance path **101**. The folding blade **103** has a swing support shaft **103***a* and a blade portion **103***b*. The swing support shaft **103***a* and the blade portion **103***b* are located, as seen from the sheet width direction (the depth direction of the plane of FIG. **3**) orthogonal to the sheet conveyance direction, at positions opposite each other across a sheet conveyance region of the first sheet conveyance path **101**, the sheet conveyance region being downstream of the folding branching portion **101***a*.

The swing support shaft **103***a* extends in a direction parallel to the sheet width direction. The folding blade **103** swings about an axis of the swing support shaft **103***a* clockwise or counterclockwise in FIG. **3**. To be more specific, the folding blade **103** is swingable so as to be positioned at a first position (see FIG. **3**) for guiding a sheet conveyed on the first sheet conveyance path **101** to a downstream part of the first sheet conveyance path **101** in the sheet conveyance direction and a second position (see FIG. **6**) for guiding a sheet conveyed on the first sheet conveyance path **101** to the second sheet conveyance path **102**.

Further, the sheet folding portion **100** includes a pair of first conveyance rollers **104** and a pair of second conveyance rollers **105**.

The pair of first conveyance rollers **104** are disposed, with respect to the sheet conveyance direction along the first sheet conveyance path **101**, at a downstream part of the first branching portion **2***a*. The pair of first conveyance rollers **104** convey a sheet on the first sheet conveyance path **101** in a direction away from the first branching portion **2***a*. The pair of first conveyance rollers **104** are constituted of a first roller **106***a* and a second roller **106***b*. The first roller **106***a* and the second roller **106***b* are disposed opposite each other across the sheet conveyance region of the first sheet conveyance path **101**, and thereby form a first nip portion **N1**.

The first switching guide **2***c* is disposed opposite the first nip portion **N1** across a sheet conveyance region of the sheet introduction path **2**. The first switching guide **2***c* is reciprocatable in directions toward and away from the first nip portion **N1**.

The first switching guide **2***c* approaches the first nip portion **N1** before a sheet conveyed on the sheet introduction path **2** reaches the first branching portion **2***a*, thereby to guide the sheet to the first nip portion **N1**, thus switching the conveyance direction of the sheet to direct the sheet into the first sheet conveyance path **101** (the third sheet discharge path **8**). Further, the first switching guide **2***c* can press such part of a sheet conveyed on the sheet introduction path **2** as has stopped over the first branching portion **2***a* and corresponds to a fold toward the first nip portion **N1**, and thus can also be used as a folding blade that performs folding processing on the sheet.

The second roller **106***b* is also used as one of the rollers that constitute one of the pairs of feed rollers **2***r* on the sheet introduction path **2**.

The pair of second conveyance rollers **105** are disposed in a downstream part of the folding branching portion **101***a* with respect to the sheet conveyance direction along the second sheet conveyance path **102**. The pair of second conveyance rollers **105** convey a sheet on the second sheet conveyance path **102** in a direction away from the folding branching portion **101***a*. The pair of second conveyance rollers **105** are constituted of the first roller **106***a* and a third roller **106***c*. The first roller **106***a* and the third roller **106***c* are disposed opposite each other across a sheet conveyance region of the second sheet conveyance path **102**, and thereby form a second nip portion **N2**.

The folding blade **103** is disposed opposite the second nip portion **N2** across the sheet conveyance region of the first sheet conveyance path **101**. The folding blade **103** is reciprocatable in directions toward and away from the second nip portion **N2**.

The folding blade **103** approaches the second nip portion **N2** before a sheet conveyed on the first sheet conveyance path **101** reaches the folding branching portion **101***a*, thereby to guide the sheet to the second nip portion **N2**, thus switching the conveyance direction of the sheet to direct the sheet into the second sheet conveyance path **102**. Further, the folding blade **103** presses, toward the second nip portion **N2**, such part of a sheet conveyed on the first sheet conveyance path **101** as has stopped over the folding branching portion **101***a* and corresponds to a fold, and thereby performs folding processing on the sheet to form a fold therein.

Next, an operation of the sheet folding portion **100** will be described with reference to FIGS. **4**, **5**, and **6**. FIGS. **4**, **5**, and **6** are sectional front views showing the sheet folding portion **100** and its vicinity shown in FIG. **3**, respectively illustrating a first stage, a second stage, and a third stage in the course of inward triple-folding processing for a sheet S. The following description of the operation of the sheet folding portion **100** deals with, as an example, folding processing for inward triple folding as shown in FIG. **2**C.

As shown in FIG. **4**, when a sheet S is introduced through the sheet introduction port **2***e* (see FIG. **1**) into the sheet introduction path **2**, a downstream part of the sheet S in the sheet conveyance direction is guided via the second branching portion **2***b* to the second sheet discharge path **6**. The second switching guide **2***d* of the second branching portion **2***b* switches the conveyance direction of the sheet S conveyed on the sheet introduction path **2** from the sheet introduction port **2***e* so as to guide the sheet S to the second sheet discharge path **6**.

The first switching guide **2***c* of the first branching portion **2***a* is retreated from the sheet introduction path **2** in a direction away from the first nip portion N**1**, that is, to a position above the sheet introduction path **2** in FIG. **4**.

Subsequently, when part of the sheet S corresponding to a first fold F**1** (see FIG. **2**C) reaches the first branching portion **2***a*, the pairs of feed rollers **2***r* in the sheet introduction path **2** and the pairs of feed rollers **6***r* in the second sheet discharge path **6** are caused to stop rotating, so that the conveyance of the sheet S is stopped. Then, such one of the pairs of feed rollers **2***r* as is disposed, in the sheet introduction path **2**, downstream of the first branching portion **2***a* (to the left of the first branching portion **2***a* in FIG. **4**) in the sheet conveyance direction, and the pairs of the feed rollers **6***r* in the second sheet discharge path **6**, are caused to rotate reversely. Thereby, such part of the sheet S as is located downstream of the first branching portion **2***a* in the sheet conveyance direction moves upstream (rightward in FIG. **4**), so that the sheet S is caused to sag at the first branching portion **2***a*.

Subsequently, the first switching guide **2***c* is moved in a direction toward the first nip portion N**1** of the pair of first conveyance rollers **104**, and makes contact with the sheet S. By making contact with the sheet S, the first switching guide **2***c* guides the sagging part of the sheet S to the first nip portion N**1**. Then, passing through the first nip portion N**1**, the sheet S has the first fold F**1** formed therein as shown in FIG. **5**.

The timing for forming the first fold F**1** in the sheet S is determined in accordance with the timing with which a sheet sensor (unillustrated) detects, in the first sheet introduction path **2**, a downstream end of the sheet S in the sheet conveyance direction, a length of the sheet S in the sheet conveyance direction, and a conveyance speed of the sheet S. This also applies to the timing for forming the second fold F**2**, which will be described later.

The folding blade **103** at the folding branching portion **101***a* is retreated from the first sheet conveyance path **101** in a direction away from the second nip portion N**2**, that is, to the right of the first sheet conveyance path **101** in FIG. **5**.

After passing through the first nip portion N**1**, the sheet S is conveyed, starting with its part where the first fold F**1** is formed, and with its two regions that extend along the sheet conveyance direction overlapped with each other, along the first sheet conveyance path **101** in a direction away from the first branching portion **2***a*. After passing through the first sheet conveyance path **101**, an upstream part of the sheet S in the conveyance direction temporarily enters the third sheet discharge path **8**.

Subsequently, when the part of the sheet S corresponding to the second fold F**2** (see FIG. **2**C) reaches the folding branching portion **101***a*, the pairs of feed rollers **2***r* in the sheet introduction path **2**, the pairs of feed rollers **6***r* in the second sheet discharge path **6**, the pair of first conveyance rollers **104**, and the pairs of feed rollers **8***r* in the third sheet discharge path **8** are caused to stop rotating, so that the conveyance of the sheet S is stopped. Then, the pairs of feed rollers **8***r* in the third sheet discharge path **8** are caused to rotate reversely. As a result, such part of the sheet S as is located downstream of the folding branching portion **101***a* in the sheet conveyance direction (as is below the folding branching portion **101***a* in FIG. **5**) moves upstream (upward in FIG. **5**), so that the sheet S is caused to sag at the folding branching portion **101***a*.

Subsequently, the folding blade **103** is moved in a direction toward the second nip portion N**2** of the pair of second conveyance rollers **105** into contact with the sheet S. By contacting the sheet S, the folding blade **103** guides the sagging part of the sheet S to the second nip portion N**2**. Then, passing through the second nip portion N**2**, the sheet S has the second fold F**2** formed therein as shown in FIG. **6**.

After passing through the second nip portion N**2**, the sheet S is conveyed, starting with its part where the second fold F**2** is formed, and with its three regions that extend along the sheet conveyance direction overlapped with each other, along the second sheet conveyance path **102** in a direction away from the pair of second conveyance rollers **105**. After passing through the second sheet conveyance path **102**, the sheet S enters the second sheet discharge path **6** via the second branching portion **2***b*, and is guided toward the second sheet discharge portion **5**. Meanwhile, at the second branching portion **2***b*, the second switching guide **2***d* switches the conveyance direction of the sheet S having passed through the second sheet conveyance path **102** and reached the second branching portion **2***b* such that the sheet S is guided to the second sheet discharge path **6**. The sheet S having been subjected to the folding processing is guided to the second sheet discharge path **6**, to be discharged through the second discharge port **51** by the pair of second discharge rollers **52** to be stacked on the second discharge tray **53** (see FIG. **1**).

Z-folding processing for Z-folding a sheet S (see FIG. **2**A) and in outward triple-folding processing for outward triple-folding a sheet S (see FIG. **2**B) can also be performed likewise, by changing the timing for forming the first fold F**1** and for forming the second fold F**2** in the sheet S, following the procedure described with reference to FIGS. **4**, **5**, and **6**. Further, double-folding processing, quadruple-folding processing, etc. can also be performed likewise on a sheet S by changing the number of times of forming a fold and the timing for forming a fold.

Next, a configuration of the second sheet discharge portion **5** and its vicinity will be described with reference to FIGS. **7**, **8**. **9** and **10**. FIG. **7** is a partial sectional front view showing the second sheet discharge portion **5** of the sheet post-processing device **1** and its vicinity shown in FIG. **1**. FIG. **8** is a perspective view showing a first sheet pressing member **54**, a second sheet pressing member **55** and a holding member **56** of the second sheet discharge portion **5** shown in FIG. **7**. FIG. **9** and FIG. **10** are partial sectional front views showing the second discharge portion **5** and its vicinity shown in FIG. **7**, with the first sheet pressing member **54** and the second sheet pressing member **55** each positioned at a retreat position. Note that, in FIG. **7**, a discharge route Re for a sheet from the second discharge port **51** is depicted.

The second sheet discharge portion **5** has, in addition to the second discharge port **51**, the pair of second discharge rollers **52**, and the second discharge tray **53**, the first sheet pressing member **54**, the second sheet pressing member **55**, and the holding member **56**.

The second discharge tray **53** is connected to the side surface **1***a* of the sheet post-processing device **1**, below the second discharge port **51**, and extends downstream in a sheet discharge direction in which a sheet S is discharged through the second discharge port **51**. The second discharge tray **53** has an upward inclination toward the downstream side in the sheet discharge direction, that is, away from the side surface

1*a* of the sheet post-processing device **1**. With this configuration, a sheet discharged onto the second discharge tray **53** is gravitationally stacked so as to be close to the side surface **1***a* of the sheet post-processing device **1**, which is on an upstream side in the sheet discharge direction.

The first sheet pressing member **54** is constituted of a sheet-shaped member, for example, having one end part (a base end part) thereof in the sheet discharge direction attached above the pair of second discharge rollers **52**, and the other end (a leading end part) thereof in the sheet discharge direction extending parallel to the sheet discharge direction. The other end part (the leading end part) of the first sheet pressing member **54** in the sheet discharging direction extends to below the pair of second discharge rollers **52**, so as to face the second discharge tray **53** across a predetermined space above the second discharge tray **53**. The first sheet pressing member **54** contacts a downstream end of a sheet in the sheet discharge direction when the sheet is discharged through the second discharge port **51**, and presses an upstream part, in the sheet discharge direction, of the sheet stacked on the second discharge tray **53**.

The first sheet pressing member **54** is attached above the pair of second discharge rollers **52** so as to be swingable about a swing shaft **54***a* extending in the sheet width direction Dw. The first sheet pressing member **54** is swingable by being contacted by a sheet. That is, the first sheet pressing member **54** swings corresponding to the number of sheets stacked on the second discharge tray **53**.

The second sheet pressing member **55** is constituted of, for example, a rod-shaped member having one end part (a base end part) thereof in the sheet discharge direction attached above the pair of second discharge rollers **52**, and extending parallel to the sheet discharge direction. The other end part (a leading end part) of the second sheet pressing member **55** in the sheet discharge direction extends to below the pair of second discharge rollers **52**, so as to, on a downstream side of the first sheet pressing member **54** in the sheet discharge direction, face the second discharge tray **53** across a predetermined space above the second discharge tray **53**.

The second sheet pressing member **55** contacts a sheet discharged through the second discharge port **51**, and presses a downstream part, in the sheet discharge direction, of the sheet stacked on the second discharge tray **53**.

The second sheet pressing member **55** is attached above the pair of second discharge rollers **52** so as to be swingable about a swing shaft **55***a* extending in the sheet width direction Dw. The second sheet pressing member **55** is swingable by being contacted by a sheet. That is, the second sheet pressing member **55** swings corresponding to the number of sheets stacked on the second discharge tray **53**.

In a case of discharging a folded sheet, having been subjected to the folding processing, into the second sheet discharge portion **5**, as shown in FIG. **7**, on a downstream side of the pair of second discharge rollers **52** in a discharge direction of the folded sheet, the first sheet pressing member **54** is positioned at a first pressing position, intersecting a discharge route Re of the folded sheet and extending downward from above the pair of second discharge rollers **52** toward the second discharge tray **53** so as to press the folded sheet. Further, in the case of discharging the folded sheet, having been subjected to the folding processing, into the second sheet discharge portion **5**, as shown in FIG. **7**, on the downstream side of the pair of second discharge rollers **52** in the discharge direction of the folded sheet, the second sheet pressing member **55** is positioned at a second pressing position, intersecting the discharge route Re of the folded sheet and extending downward from above the pair of second discharge rollers **52** toward the second discharge tray **53** so as to press the folded sheet.

On the other hand, in a case of discharging an unfolded sheet, not having been subjected to the folding processing, into the second sheet discharge portion **5**, as shown in FIG. **9**, on the downstream side of the pair of second discharge rollers **52** in the discharge direction of the folded sheet, the first sheet pressing member **54** is positioned at a first retreat position retreated to above the discharge route Re (see FIG. **7**) of a folded sheet. Further, in the case of discharging the unfolded sheet, not having been subjected to the folding processing, into the second sheet discharge portion **5**, as shown in FIG. **10**, on the downstream side of the pair of second discharge rollers **52** in the discharge direction of the folded sheet, the second sheet pressing member **55** is positioned at a second retreat position retreated to above the discharge route Re (see FIG. **7**) of a folded sheet.

As described above, the first sheet pressing member **54** is swingable to the first pressing position and to the first retreat position. The second sheet pressing member **55** is swingable to the second pressing position and to the second retreat position.

The holding member **56** is disposed on an upper surface of the sheet post-processing device **1**. The holding member **56** is constituted of a hollow box having a substantially rectangular-parallelepiped shape extending in the sheet width direction Dw. The holding member **56** holds the first sheet pressing member **54** and the second sheet pressing member **55** so as to be swingable about the swing shafts **54***a* and **55***a*, respectively, which each extend in the sheet width direction Dw. A side surface **56***a* of the holding member **56** faces the same direction as a side surface **1***a* of the sheet post-processing device **1** does, the second discharge tray **53** being located on the side surface **1***a*, and the first sheet pressing member **54** and the second sheet pressing member **55** protrude toward the second discharge tray **53**. The holding member **56** has a first holding portion **561** and a second holding portion **562**.

The first holding portion **561** holds the first sheet pressing member **54**. To be more specific, the first holding portion **561** selectively holds the first sheet pressing member **54** at the first pressing position and the first retreat position (see FIG. **9**). Further, the second holding portion **562** holds the second sheet pressing member **55**. To be more specific, the second holding portion **562** selectively holds the second sheet pressing member **55** at the second pressing position and the second retreat position (see FIG. **10**). The selective positioning with respect to the first sheet pressing member **54** and the second sheet pressing member **55** is manually done by a user.

According to the above configuration, in a case where, for example, sheets with low stiffness (hardness, firmness) are to be discharged, the first sheet pressing member **54** and the second sheet pressing member **55** can be positioned at their respective retreat positions. Thereby, it becomes possible, regardless of the type of sheets, to have a large number of sheets stacked on the second discharge tray **53** in an orderly manner.

Next, detailed configurations of the first sheet pressing member **54**, the second sheet pressing member **55**, and the holding member **56** will be described.

FIG. **11** is a top view and FIG. **12** is a perspective view as seen from below, of the holding member **56** of the second sheet discharge portion **5** shown in FIG. **7**. FIG. **13** is a side view of a first protruding portion **5613***a* of the holding member **56** shown in FIG. **12**, as seen from the sheet discharge direction. As mentioned previously, the holding member **56** has the first holding portion **561** and the second holding portion **562**.

The first holding portion **561** is formed inside the holding member **56**. The first holding portion **561** includes a pair of first holding portions **561** disposed one on each of opposite outer sides of a central part of the holding member **56** in the sheet width direction Dw orthogonal to the sheet discharge direction. The first holding portion **561** has a fitting recessed portion **5611**, a first bearing portion **5612**, and a locking portion **5613**.

The fitting recessed portion **5611** is a portion recessed upward from a lower side of the holding member **56**. On a downstream side of the fitting recessed portion **5611** in the sheet discharge direction, an opening portion **56***b* is disposed. The fitting recessed portion **5611** has a space to accommodate an upstream part of the first sheet pressing member **54** in the sheet discharge direction. In other words, the fitting recessed portion **5611** has a space that allows swinging of a later-described locking piece **542** of the first sheet pressing member **54**.

The first bearing portion **5612** is formed in a plate-shaped wall portion **561***a* of the fitting recessed portion **5611**, the plate-shaped wall portion **561***a* extending in an up-down direction and the sheet discharge direction. The first bearing portion **5612** is constituted as a hole portion that penetrates the plate-shaped wall portion **561***a*. The first bearing portion **5612** is formed at each of two positions aligned in the sheet width direction Dw corresponding to two swing shafts **54***a* (see FIG. **14**) of the first sheet pressing member **54**. The first bearing portion **5612** has inserted therein the swing shaft **54***a* of the first sheet pressing member **54**. The first sheet pressing member **54** is swingably connected to the first holding portion **561** via the swing shaft **54***a*. In other words, the first bearing portion **5612** swingably supports the swing shaft **54***a* of the first sheet pressing member **54**.

The locking portion **5613** is provided in the fitting recessed portion **5611**. The locking portion **5613** contacts the later-described locking piece **542** of the first sheet pressing member **54** so as to restrict swinging of the first sheet pressing member **54** to a predetermined position. The locking portion **5613** has the first protruding portion **5613***a*.

The first protruding portion **5613***a* is formed on a side surface **561***b* of the plate-shaped wall portion **561***a* of the fitting recessed portion **5611**, the plate-shaped wall portion **561***a* extending in the up-down direction and the sheet discharge direction. The first protruding portion **5613***a* is disposed closer to the center than the first bearing portion **5612** with respect to the sheet width direction Dw, and protrudes outward in the sheet width direction Dw so as to face the first bearing portion **5612**. That is, the first protruding portion **5613***a* protrudes from the side surface **561***b* of the fitting recessed portion **5611**. In other words, the first protruding portion **5613***a* protrudes toward a direction in which the first sheet pressing member **54** attached via the first bearing portion **5612** is located. Further, to be more specific, the first protruding portion **5613***a* protrudes, toward the first sheet pressing member **54**, so as to intersect a swing trajectory of the first sheet pressing member **54**.

FIG. **14** is a perspective view of the first sheet pressing member **54** of the second sheet discharge portion **5** shown in FIG. **7**. The first sheet pressing member **54** has a connection portion **541**, the locking piece **542**, a pressing portion **543**, and a handle portion **544**.

The connection portion **541** is disposed on a side of the one end part (the base end part) of the first sheet pressing member **54** in the sheet discharge direction. The connection

portion **541** includes two connection portions **541** formed one at each of two positions along the sheet width direction Dw so as to be coupled by the handle portion **544** extending in the sheet width direction Dw. The connection portion **541** and the handle portion **544** are integrally formed of a synthetic resin, for example. At each of opposite end parts of the connection portion **541** in the sheet width direction Dw, the swing shaft **54***a* is disposed so as to extend outward in the sheet width direction Dw. A downstream part of the connection portion **541** in the sheet discharge direction protrudes through the opening portion **56***b* provided in the side surface **56***a* of the holding member **56** toward an outside of the holding member **56** (see FIG. **8**).

The locking piece **542** extends from the swing shaft **54***a* toward one side in a direction that is parallel to the sheet discharge direction. In other words, the locking piece **542** is provided on a side that is opposite, with respect to the swing shaft **54***a*, to the side on which the pressing portion **543** is provided. The locking piece **542** engages with the locking portion **5613** of the first holding portion **561** when the first sheet pressing member **54** is positioned at the first pressing position and the first retreat position. As to a detailed configuration of the locking piece **542**, a description will be given later with reference to FIG. **15**, FIG. **16**, and FIG. **17**.

The pressing portion **543** is disposed on a side of the other end part (the leading end part) of the first sheet pressing member **54** in the sheet discharge direction. In other words, the pressing portion **543** extends from the swing shaft **54***a* toward the other side in the direction that is parallel to the sheet discharge direction. The pressing portion **543** includes two pressing portions **543** fixed one to each of the two connection portions **541**, such that the two pressing portions **543** are disposed at two positions along the sheet width direction Dw so as to be parallel to each other. The pressing portion **543** is constituted of a sheet-shaped member, for example, that extends parallel to the sheet discharge direction. The pressing portion **543** contacts a folded sheet discharged through the second discharge port **51**, and presses an upstream part of the folded sheet stacked on the second discharge tray **53** in the discharge direction of the folded sheet.

The handle portion **544** is disposed downstream of the connection portion **541** in the sheet discharge direction, and is located outside the holding member **56**. The handle portion **544** extends in the sheet width direction Dw, and couples the two connection portions **541** disposed at the two positions along the sheet width direction Dw. The handle portion **544** is integrally formed with the connection portion **541**, for example. A user can hold the handle portion **544** by hand to displace the first sheet pressing member **54**.

FIG. **15** is a partial rear view of the first sheet pressing member **54** shown in FIG. **14**. FIG. **16** is a partial sectional front view of the first sheet pressing member **54** (the first pressing position) and the holding member **56** of the second sheet discharge portion **5** shown in FIG. **7**. FIG. **17** is a partial sectional front view of the first sheet pressing member **54** (the first retreat position) and the holding member **56** of the second sheet discharge portion **5** shown in FIG. **7**.

The locking piece **542** is provided on a side opposite to the pressing portion **543** across the swing shaft **54***a* with respect to the sheet discharge direction. The locking piece **542** engages with the first protruding portion **5613***a* of the holding member **56**. The locking piece **542** has a lower contact portion **542**L and an upper contact portion **542**U.

The lower contact portion **542**L is, as shown in FIG. **16**, constituted of a flat-surface portion that extends substantially horizontally when the first sheet pressing member **54** is in the state (position) of being held at the first pressing position. The lower contact portion **542**L contacts the first protruding portion **5613***a* from above. That is, the first protruding portion **5613***a* engages with the locking piece **542** of the first sheet pressing member **54** from one side (a lower side) in the up-down direction to hold the first sheet pressing member **54** at the first pressing position. In other words, the locking portion **5613** locks the locking piece **542** when the first sheet pressing member **54** is positioned at the first pressing position, and holds the first sheet pressing member **54** at the first pressing position.

The upper contact portion **542**U is, as shown in FIG. **17**, constituted of a flat-surface portion that extends substantially horizontally when the first sheet pressing member **54** in the state (position) of being held at the first retreat position. The upper contact portion **542**U contacts the first protruding portion **5613***a* from below. That is, the first protruding portion **5613***a* engages with the locking piece **542** of the first sheet pressing member **54** from the other side (an upper side) in the up-down direction to hold the first sheet pressing member **54** at the first retreat position. In other words, the locking portion **5613** locks the locking piece **542** when the first sheet pressing member **54** is positioned at the first retreat position, and holds the first sheet pressing member **54** at the first retreat position.

According to the above configuration, by using the single first protruding portion **5613***a*, it is possible to hold the first sheet pressing member **54** at the two positions, namely, the first pressing position and the first retreat position. That is, it is possible to form the first sheet pressing member **54** and the holding member **56** with a simple configuration. This makes it possible, with a simple configuration, to stack a large number of sheets on the second discharge tray **53** in an orderly manner.

To be more specific, when the first sheet pressing member **54** is in the state (position) of being held at the first pressing position as shown in FIG. **16**, the first sheet pressing member **54** has an upstream end part thereof in the sheet discharge direction in contact with an upper surface **56***c* of the holding member **56** (the fitting recessed portion **5611**). As a result, the first sheet pressing member **54** is prevented from rotating counterclockwise in FIG. **16**. The locking portion **5613** contacts the locking piece **542** to restrict swinging of the first sheet pressing member **54** to a predetermined position.

That is, as shown in FIG. **16**, the first protruding portion **5613***a* contacts the locking piece **542** from below such that the locking piece **542** is sandwiched between an upper end of the first protruding portion **5613***a* and the upper surface **56***c* of the fitting recessed portion **5611**, and thereby holds the first sheet pressing member **54** at the first pressing position. Further, as shown in FIG. **17**, the locking piece **542** engages with a lower end of the first protruding portion **5613***a*, and thereby holds the first sheet pressing member **54** at the first retreat position.

According to the above configuration, it is possible to suppress unnecessarily close approach of the first sheet pressing member **54** to the second discharge port **51** when the first sheet pressing member **54** is positioned at the first pressing position. Thus, it is possible to prevent obstruction to the conveyance of a sheet to be discharged onto the second discharge tray **53** through the second discharge port **51**. That is, it becomes possible to stack sheets on the second discharge tray **53** in an orderly manner.

Furthermore, as described above, the locking piece **542** is provided on the opposite side of the pressing portion **543** with respect to the sheet discharge direction across the swing shaft **54***a*. That is, by using a vacant area (space) upstream of the swing shaft **54***a* with respect to the sheet discharge direction, it is possible to provide a point of engagement of the first sheet pressing member **54** with the holding member **56**.

Further, referring back to FIG. **13**, the first protruding portion **5613***a* has a leading end surface **5613**E, an upper inclined surface **5613**U, and a lower inclined surface **5613**L. The upper inclined surface **5613**U and the lower inclined surface **5613**L are formed at parts at which the first protruding portion **5613***a* contacts the locking piece **542**.

The leading end surface **5613**E is formed at a leading end part of the first protruding portion **5613***a* protruding from the side surface **561***b* of the fitting recessed portion **5611**. The leading end surface **5613**E extends along the side surface **561***b* in the up-down direction and the sheet discharge direction. In other words, the leading end surface **5613**E extends substantially parallel to the side surface **561***b*.

The upper inclined surface **5613**U is formed at an upper part of the first protruding portion **5613***a*. The upper inclined surface **5613**U has an inclination such that the upper inclined surface **5613**U protrudes toward the first sheet pressing member **54** downward. In other words, the upper inclined surface **5613**U is inclined downward from an upper side of the first protruding portion **5613***a* toward the leading end surface **5613**E into contact with the leading end surface **5613**E. When the first sheet pressing member **54** is displaced from the first retreat position to the first pressing position, the upper inclined surface **5613**U contacts the lower contact portion **542**L of the locking piece **542** from below. The upper inclined surface **5613**U guides movement of the locking piece **542**.

The lower inclined surface **5613**L is formed at a lower part of the first protruding portion **5613***a*. The lower inclined surface **5613**L has an inclination such that the lower inclined surface **5613**U protrudes toward the first sheet pressing member **54** upward. In other words, the lower inclined surface **5613**L is inclined upward from a lower side of the first protruding portion **5613***a* toward the leading end surface **5613**E into contact with the leading end surface **5613**E. When the first sheet pressing member **54** is displaced from the first pressing position to the first retreat position, the lower inclined surface **5613**L contacts the upper contact portion **542**U of the locking piece **542** from above. The lower inclined surface **5613**L guides movement of the locking piece **542**.

With respect to the up-down direction, the lower inclined surface **5613**L has a smaller inclination angle than the upper inclined surface **5613**U. According to this configuration, it is possible to displace the first sheet pressing member **54** from the first retreat position to the first pressing position with ease. On the other hand, with respect to the up-down direction, the upper inclined surface **5613**U is inclined at an angle close to a right angle. According to this configuration, it is possible to suppress unintentional displacement of the first sheet pressing member **54** positioned at the first pressing position toward the first retreat position caused by a sheet discharged toward the second discharge tray **53** contacting the pressing portion **543**.

Further, referring back to FIG. **8** and FIG. **11**, the second holding portion **562** is formed at an upper part of the holding member **56**. The second holding portion **562** includes a pair of second holding portions **562** disposed one on each of opposite outer sides of the first holding portion **561** with respect to the sheet width direction Dw. By disposing the pair of first holding portions **561** one on each side toward the center in the sheet width direction Dw, and disposing the pair of second holding portions **562** one on each of the opposite outer sides of the pair of first holding portions **561** with respect to the sheet width direction Dw, it is possible to suppress contacting of the first sheet pressing member **54** and the second sheet pressing member **55** with each other when they swing.

The second holding portion **562** has a groove portion **5621**, a second bearing portion **5622**, and a second protruding portion **5623**.

The groove portion **5621** includes two groove portions **5621** disposed one at each of two positions located on opposite outer sides of the pair of first holding portions **561** (the first sheet pressing member **54**) in the sheet width direction Dw, and extending along the sheet discharge direction. The groove portion **5621** is formed as a recessed portion that is recessed downward. The groove portion **5621** has a groove width (an interval in the sheet width direction Dw) that allows free displacement of the second sheet pressing member **55** when the second sheet pressing member **55**, which is constituted of a rod-shaped member, swings.

The second bearing portion **5622** is formed in a plate-shaped wall portion **562***a* inside the groove portion **5621**, the plate-shaped wall portion **562***a* extending in the up-down direction and the sheet discharge direction. The second bearing portion **5622** is formed as a hole portion that penetrates through the plate-shaped wall portion **562***a*. The second bearing portion **5622** is formed at each of two positions along the sheet width direction Dw corresponding to two swing shafts **55***a* that the swing shaft **55***a* (see FIG. **19**) of the second sheet pressing member **55** includes. The second bearing portion **5622** has the swing shaft **55***a* of the second sheet pressing member **55** inserted therein. That is, the second sheet pressing member **55** is swingably coupled to the second holding portion **562** via the swing shaft **55***a*.

FIG. **18** is a partial side view showing the second holding portion **562** of the holding member **56** of the second sheet discharge portion **5** and its vicinity shown in FIG. **7**. The second protruding portion **5623** is disposed in one of the two groove portions **5621**. The second protruding portion **5623** is formed in a plate-shaped wall portion **562***b* inside the groove portion **5621**, the plate-shaped wall portion **562***b* extending in the up-down direction and the sheet discharge direction. The second protruding portion **5623** is disposed above with respect to a bottom part of the groove portion **5621**.

The second protruding portion **5623** protrudes toward an inside of the groove portion **5621**. In other words, the second protruding portion **5623** protrudes toward where the second sheet pressing member **55** attached via the second bearing portion **5622** is located. To be more specific, the second protruding portion **5623** protrudes toward the second sheet pressing member **55** so as to intersect a swing trajectory of the second sheet pressing member **55**.

FIG. **19** is a perspective view of the second sheet pressing member **55** of the second sheet discharge portion **5** shown in FIG. **7**. In the present embodiment, the second sheet pressing member **55** is constituted of a single rod-shaped member bent at a plurality of predetermined positions. The second sheet pressing member **55** has a connection portion **551**, a pressing portion **552**, and a handle portion **553**.

The connection portion **551** is disposed on a side of one end part (a base end part) of the second sheet pressing member **55** in the sheet discharge direction. The connection portion **551** includes two connection portions **551** formed at opposite end parts of the rod-shaped member, one at each of two positions along the sheet width direction Dw. At each of the connection portion **551**, the swing shaft **55***a* is disposed which is bent in the sheet width direction Dw with respect to the sheet discharge direction so as to extend outward in the sheet width direction Dw.

The pressing portion **552** is disposed on a side of the other end part (a leading end part) of the second sheet pressing member **55** in the sheet discharge direction. The pressing portion **552** includes two pressing portions **552** which each extend continuously from a corresponding one of the two connection portions **551** in the sheet discharge direction so as to be disposed parallel to each other at two positions along the sheet width direction Dw. The two pressing portions **552** are disposed one on each of opposite outer sides of the first sheet pressing member **54** with respect to the sheet width direction Dw. The pressing portion **552** has a downstream part thereof bent upward. The two pressing portions **552** are coupled to each other by the handle portion **553** extending in the sheet width direction Dw. The pressing portion **552** contacts a sheet discharged through the second discharge port **51**, and presses the downstream part, in the sheet discharge direction, of the sheet stacked on the second discharge tray **53**.

The handle portion **553** is disposed at the downstream end of the pressing portion **552** in the sheet discharge direction so as to be located at a central part of the second sheet pressing member **55** in the sheet width direction Dw. The handle portion **553** extends in the sheet width direction Dw, and couples the pressing portions **552** disposed at the two positions along the sheet width direction Dw. The user can hold the handle portion **553** by hand to displace the second sheet pressing member **55**.

The groove portion **5621**, which extends along the sheet discharge direction, has the second sheet pressing member **55** placed thereon toward a downstream side of the groove portion **5621** in the sheet discharge direction, and thereby holds the second sheet pressing member **55** at the second pressing position (see FIG. **9**). The second protruding portion **5623**, which is disposed above with respect to the bottom part of the groove portion **5621**, has the second sheet pressing member **55** placed thereon so as to be hooked on an upper side thereof, and thereby holds the second sheet pressing member **55** at the second retreat position (see FIG. **10**).

According to the above configuration, by using the groove portion **5621** and the second protruding portion **5623**, it is possible to hold the second sheet pressing member **55** at the two positions, namely, the second pressing position and the second retreat position. That is, it is possible to form the second sheet pressing member **55** and the holding member **56** with a simple configuration. This makes it possible, with a simple configuration, to stack a large number of sheets on the second discharge tray **53** in an orderly manner.

Further, the second bearing portion **5622** is disposed at a middle part of the groove portion **5621** with respect to the sheet discharge direction (see FIG. **8** and FIG. **11**). In other words, the groove portion **5621** extends substantially in a straight line toward both downstream and upstream sides of the second bearing portion **5622** with respect to the sheet discharge direction.

FIG. **20** is a partial sectional front view showing the second discharge portion **5** and its vicinity shown in FIG. **7**, with the second sheet pressing member **55** positioned at a third retreat position. As shown in FIG. **20**, the second sheet pressing member **55** can be positioned at the third retreat position which is displaced, with respect to the second pressing position, to an opposite side centered about the swing shaft **55***a*. In other words, the second sheet pressing member **55** can be disposed on the upper surface of the sheet post-processing device **1**. That is, it is possible to retreat the second sheet pressing member **55** from above the second sheet discharge portion **5** to another position.

The above-described embodiment is by no means meant to limit the scope of the present disclosure, and various modifications can be made and implemented within the scope not departing from the gist of the present disclosure.

What is claimed is:

1. A sheet post-processing device, comprising:

a discharge port having a pair of discharge rollers that discharges a sheet;

a discharge tray on which the sheet discharged through the discharge port is stacked;

a first sheet pressing member that is attached above the pair of discharge rollers and presses an upstream part, in a sheet discharge direction, of the sheet stacked on the discharge tray;

a second sheet pressing member that is attached above the pair of discharge rollers and presses a downstream part, in the sheet discharge direction, of the sheet stacked on the tray; and

a holding member that holds the first sheet pressing member and the second sheet pressing member,

wherein

the holding member has

a first holding portion that holds the first sheet pressing member such that the first sheet pressing member is swingable to a first pressing position, at which the first sheet pressing member extends, intersecting a discharge route of the sheet, downward from above the pair of discharge rollers toward the discharge tray to press the sheet, and to a first retreat position retreated to above the discharge route of the sheet, and

a second holding portion that holds the second sheet pressing member such that the second sheet pressing member is swingable to a second pressing position, at which the second sheet pressing member extends, intersecting the discharge route of the sheet, downward from above the pair of discharge rollers toward the discharge tray to press the sheet, and to a second retreat position retreated to above the discharge route of the sheet,

the first holding portion selectively holds the first sheet pressing member at the first pressing position and the first retreat position, and

the second holding portion selectively holds the second sheet pressing member at the second pressing position and the second retreat position.

2. The sheet post-processing device according to claim **1**, wherein

the first sheet pressing member has

a swing shaft,

a locking piece extending from the swing shaft toward one side, and

a pressing portion extending from the swing shaft toward an other side,

the first holding portion has

a fitting recessed portion that allows swinging of the locking piece,

a bearing portion that swingably supports the swing shaft of the first sheet pressing member, and

a locking portion that is provided in the fitting recessed portion, and contacts the locking piece so as to restrict swinging of the first sheet pressing member to a predetermined position, and

the locking portion

locks the locking piece when the first sheet pressing member is positioned at the first pressing portion so as to hold the first sheet pressing member at the first pressing position, and

locks the locking piece when the first sheet pressing member is positioned at the first retreat position so as to hold the first sheet pressing member at the first retreat position.

3. The sheet post-processing device according to claim **2**, wherein

the locking portion has a first protruding portion that protrudes from a side surface of the fitting recessed portion, such that

the first sheet pressing member is held at the first pressing position by the locking piece being sandwiched between an upper end of the first protruding portion and an upper surface of the fitting recessed portion, and

the first sheet pressing member is held at the first retreat position by the locking piece being engaged with a lower end of the first protruding portion.

4. The sheet post-processing device according to claim **3**, wherein

the first protruding portion has

a leading end surface that extends along the side surface of the fitting recessed portion,

an upper inclined surface that is inclined downward from above toward the leading end surface into contact with the leading end surface, and

a lower inclined surface that is inclined upward from below toward the leading end surface into contact with the leading end surface, and

the upper inclined surface and the lower inclined surface guide movement of the locking piece.

5. The sheet post-processing device according to claim **1**, wherein

the second holding portion has

a groove portion that extends along the sheet discharge direction and holds the second sheet pressing member at the second pressing position, and

a second protruding portion that is disposed above with respect to a bottom part of the groove portion, protrudes so as to intersect a swing trajectory of the second sheet pressing member, and holds the second sheet pressing member at the second retreat position.

6. The sheet post-processing device according to claim **1**, wherein

the first sheet pressing member and the second sheet pressing member each have a handle portion.

7. The sheet post-processing device according to claim **1**, wherein

the first holding portion includes a pair of first holding portions disposed one on each of opposite outer sides of a central part of the holding member in a width direction that is orthogonal to the sheet discharge direction, and

the second holding portion includes a pair of second holding portions disposed one on each of opposite outer sides of the first holding portion with respect to the width direction.

* * * * *